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MIL-M-28787/502(SH)
29 DECEMBER 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1397 INTERFACE, TYPE B/PIC MODULE, KEY CODE AGL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-1397 interface, type B/Peripheral Input Channel (PIC) module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.918 failures per million operating hours for class I or 9.922 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AGL.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Power supply stress	IV	S	----------Module shall pass all remaining tests----------						
D1-D4	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D5	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D6-D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D9	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D10-D12	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D13,D14	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E4	Ios	VI	S	5.0	265.0	5.0	265.0	5.0	265.0	mA
E5-E36	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E37,E38	Ios	VI	S	-3.1	-1.3	-3.1	-1.3	-3.1	-1.3	mA
E39-E49	Ios	VI	S	5.0	265.0	5.0	265.0	5.0	265.0	mA
E50,E51	Ios	VI	S	-6.04	-5.24	-6.04	-5.24	-6.04	-5.24	mA
E52-E56	Ios	VI	S	5.0	265.0	5.0	265.0	5.0	265.0	mA
F1,F2	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F3	Iih	VII	X	-12.2	-2.85	-12.2	-2.85	-12.2	-2.85	mA
F4-F7	Iih	VII	X	-0.8	0.12	-0.8	0.12	-0.8	0.12	mA
F8	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F9-F12	Iih	VII	X	-0.8	0.12	-0.8	0.12	-0.8	0.12	mA
F13	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F14	Iih	VII	X	-1.6	0.08	-1.6	0.08	-1.6	0.08	mA
F15,F16	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F17	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F18	Iih	VII	X	-1.44	0.08	-1.44	0.08	-1.44	0.08	mA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 3/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F19,F20	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F21	Iih	VII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
F22	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F23	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G1,G2	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G3	Iil	VIII	X	-12.2	-2.85	-12.2	-2.85	-12.2	-2.85	mA
G4-G7	Iil	VIII	X	-0.8	0.12	-0.8	0.12	-0.8	0.12	mA
G8	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G9-G12	Iil	VIII	X	-0.8	0.12	-0.8	0.12	-0.8	0.12	mA
G13	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G14	Iil	VIII	X	-1.6	0.08	-1.6	0.08	-1.6	0.08	mA
G25,G16	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G17	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G18	Iil	VIII	X	-1.44	0.08	-1.44	0.08	-1.44	0.08	mA
G19,G20	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G21	Iil	VIII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
G22	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G23	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
H1-H4	Voh	IX	X	-0.6	0.0	-0.6	0.0	-0.6	0.0	V
H5-H38	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
H39-H49	Voh	IX	X	-0.6	0.0	-0.6	0.0	-0.6	0.0	V
H50-H54	Voh	IX	X	-0.6	0.0	-0.6	0.0	-0.6	0.0	V
J1-J4	Vol	X	X	-4.5	-2.5	-4.5	-2.5	-4.5	-2.5	V
J5-J38	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
J39-J49	Vol	X	X	-4.5	-2.5	-4.5	-2.5	-4.5	-2.5	V
J50,J51	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
J52-J56	Vol	X	X	-4.5	-2.5	-4.5	-2.5	-4.5	-2.5	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
L1-L9	Vio	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XII	S	4.4	8.2	4.4	8.2	4.4	8.2	uF
M2	PSFC	XII	S	3.8	7.0	3.8	7.0	3.8	7.0	uF
M3	PSFC	XII	S	11.0	20.2	11.0	20.2	11.0	20.2	uF
M4	PSFC	XII	S	1.4	2.6	1.4	2.6	1.4	2.6	uF
N1	Icc	XIII	X	330.0	790.0	330.0	790.0	330.0	790.0	mA
N2	Ico	XIII	X	-67.0	-23.75	-67.0	-23.75	-67.0	-23.75	mA
N3	Ico	XIII	X	-156.0	-60.0	-156.0	-60.0	-156.0	-60.0	mA
P1,P2	Functional	XIV	X	----------		See table XIV		----------		
Q1,Q2	Dynamic	XV	S	----------		See table XV		----------		
Q3,Q4	Dynamic	XV	Q	----------		See table XV		----------		
R1	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R2	Ci/o	XVII	Q	2	100	2	100	2	100	pF
R3	Ci/o	XVII	Q	8	200	8	200	8	200	pF
R4	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R5-R8	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R9	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R10	Ci/o	XVII	Q	4	175	4	175	4	175	pF
R11	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R12-R14	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R15	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R16	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R17-R24	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R25	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R26	Ci/o	XVII	Q	4	200	4	200	4	200	pF
R27	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R28	Ci/o	XVII	Q	2	100	2	100	2	100	pF
R29	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R30	Ci/o	XVII	Q	4	125	4	125	4	125	pF

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R31,R32	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R33	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R34	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R35,R36	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R37	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R38	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R39-R53	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R54	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R55	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R56-R62	Ci/o	XVII	Q	8	175	8	175	8	175	pF
R63	Ci/o	XVII	Q	8	200	8	200	8	200	pF
T1-T4	PSTA	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifing number M28787/502-1 for class I, and M28787/502-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc, (V2) of -5.72 V dc, and (V3) of -16.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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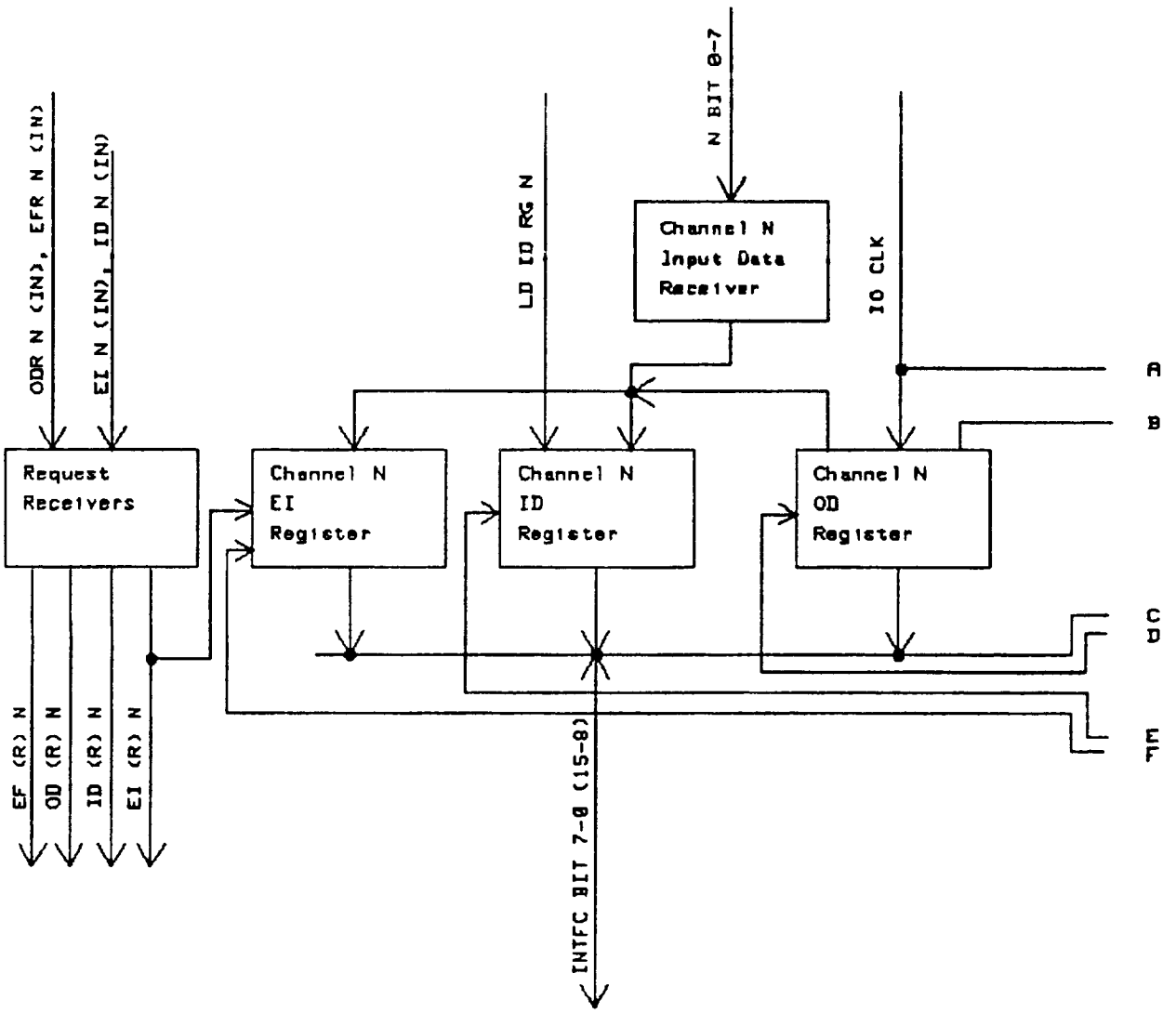


FIGURE 1. AGL functional block diagram.

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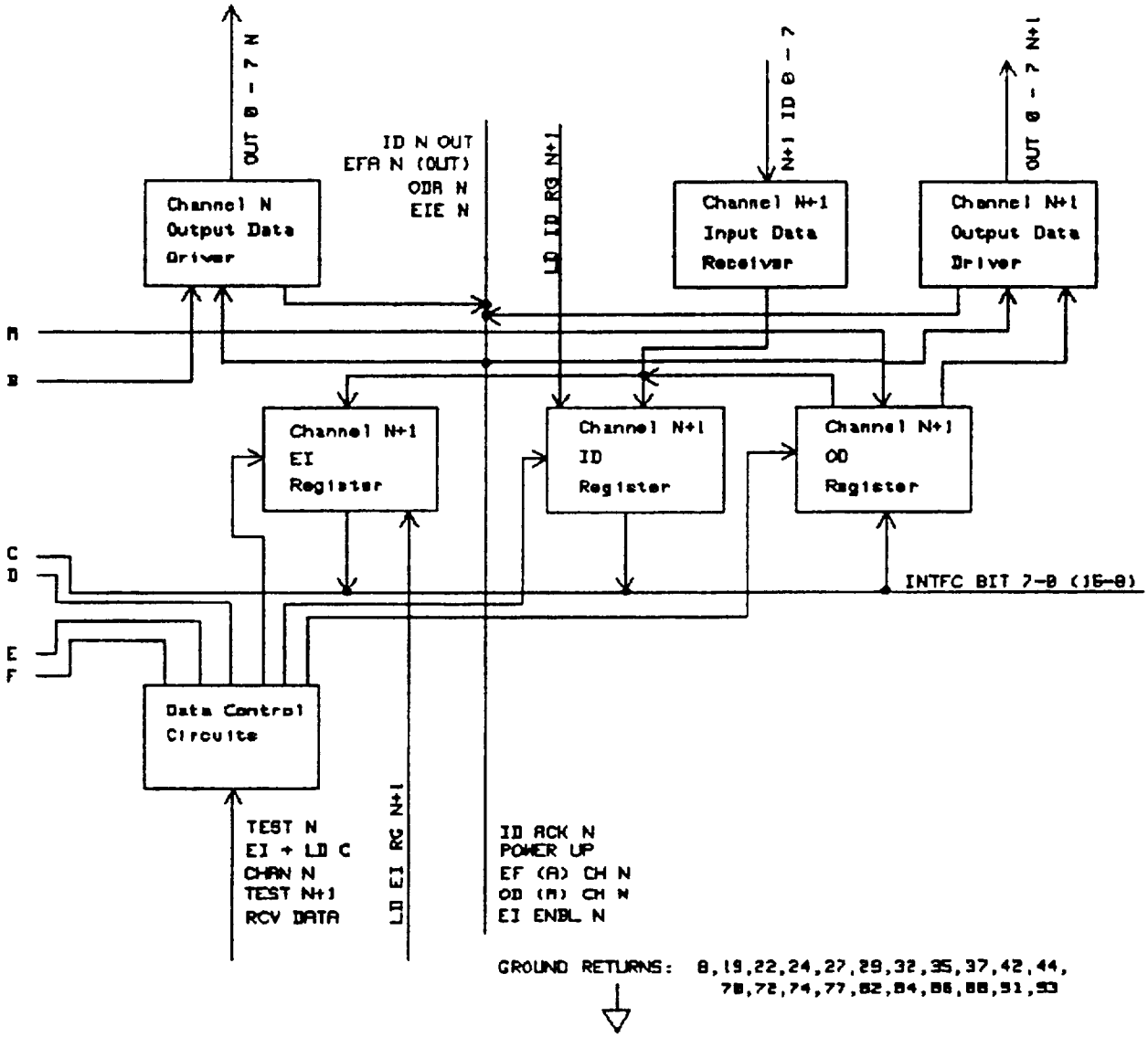


FIGURE 1. AGL functional block diagram - Continued.

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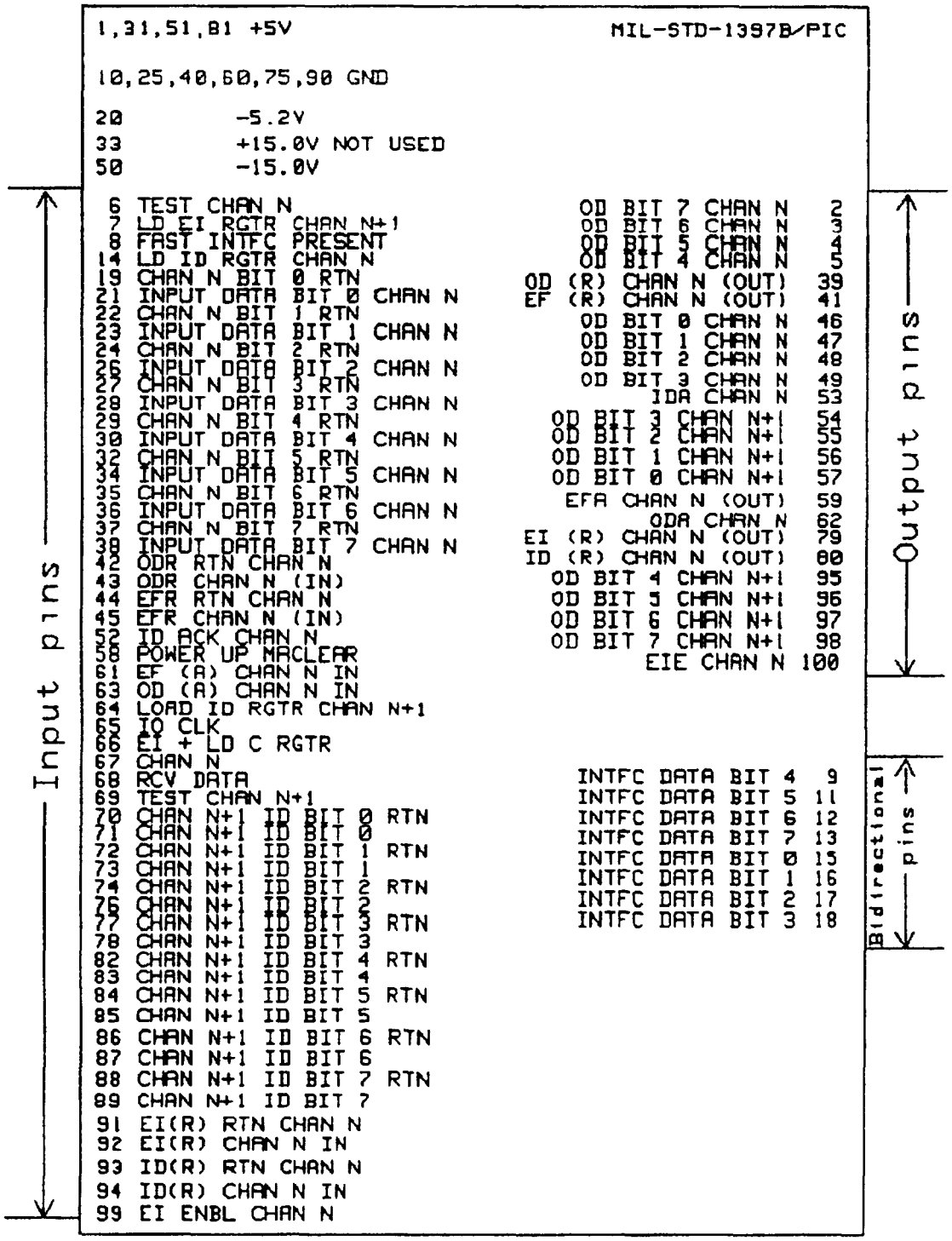


FIGURE 2. AGL pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing.. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature

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stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

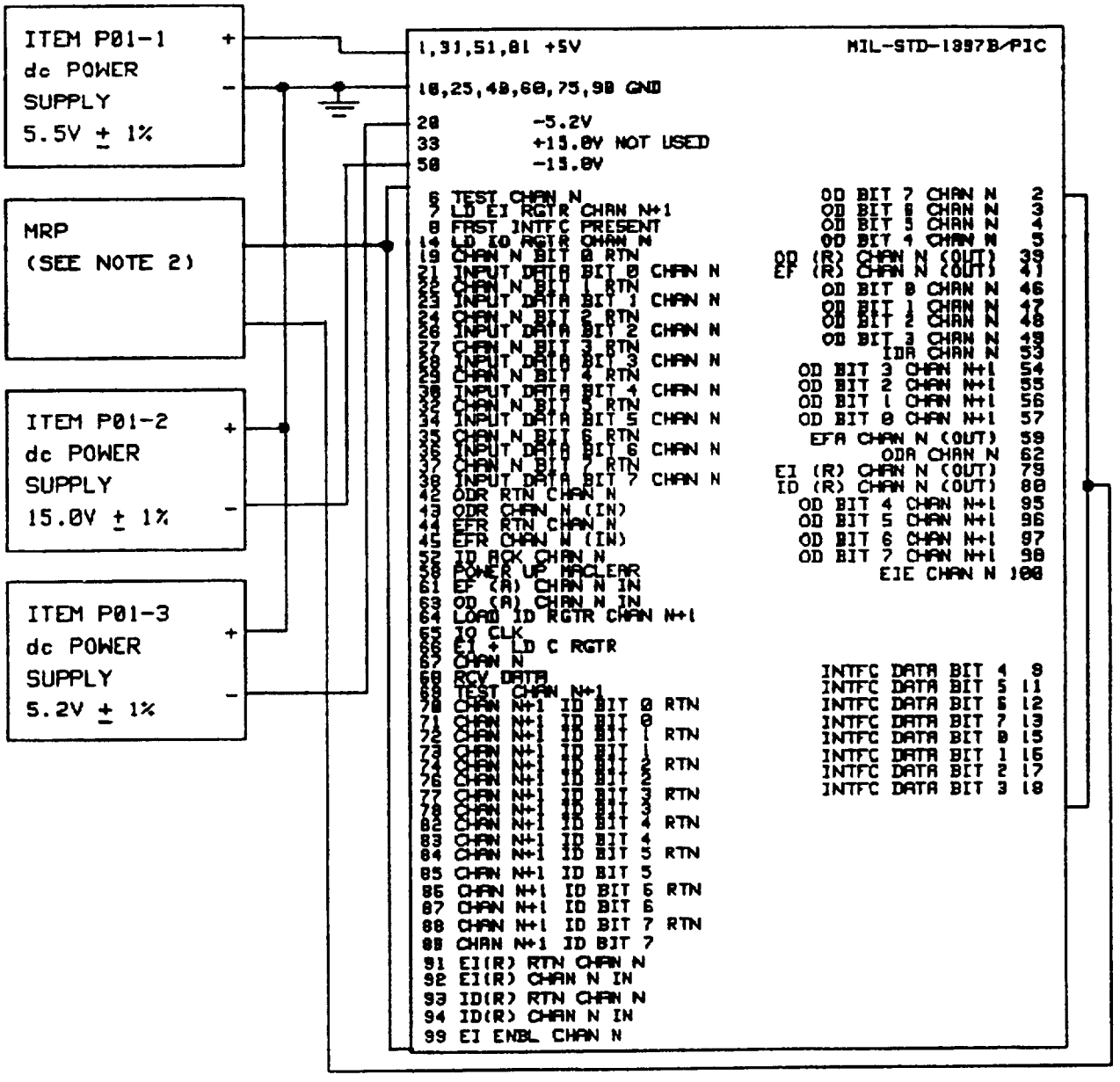
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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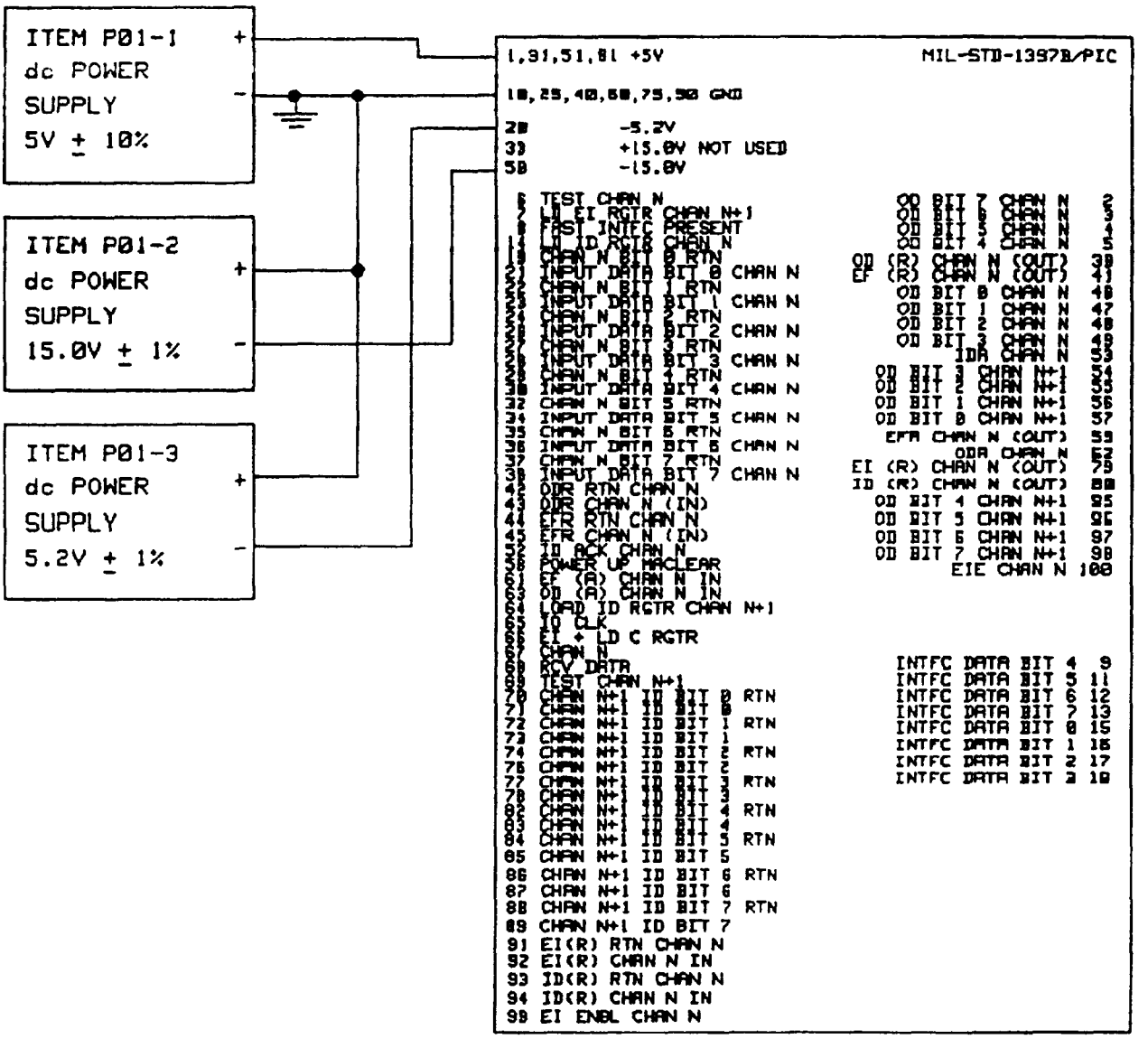
NOTES:

1. Caution: Turn the -5.2 V power supply on first and off last.

2. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AGL operating temperature and component temperature test circuit.

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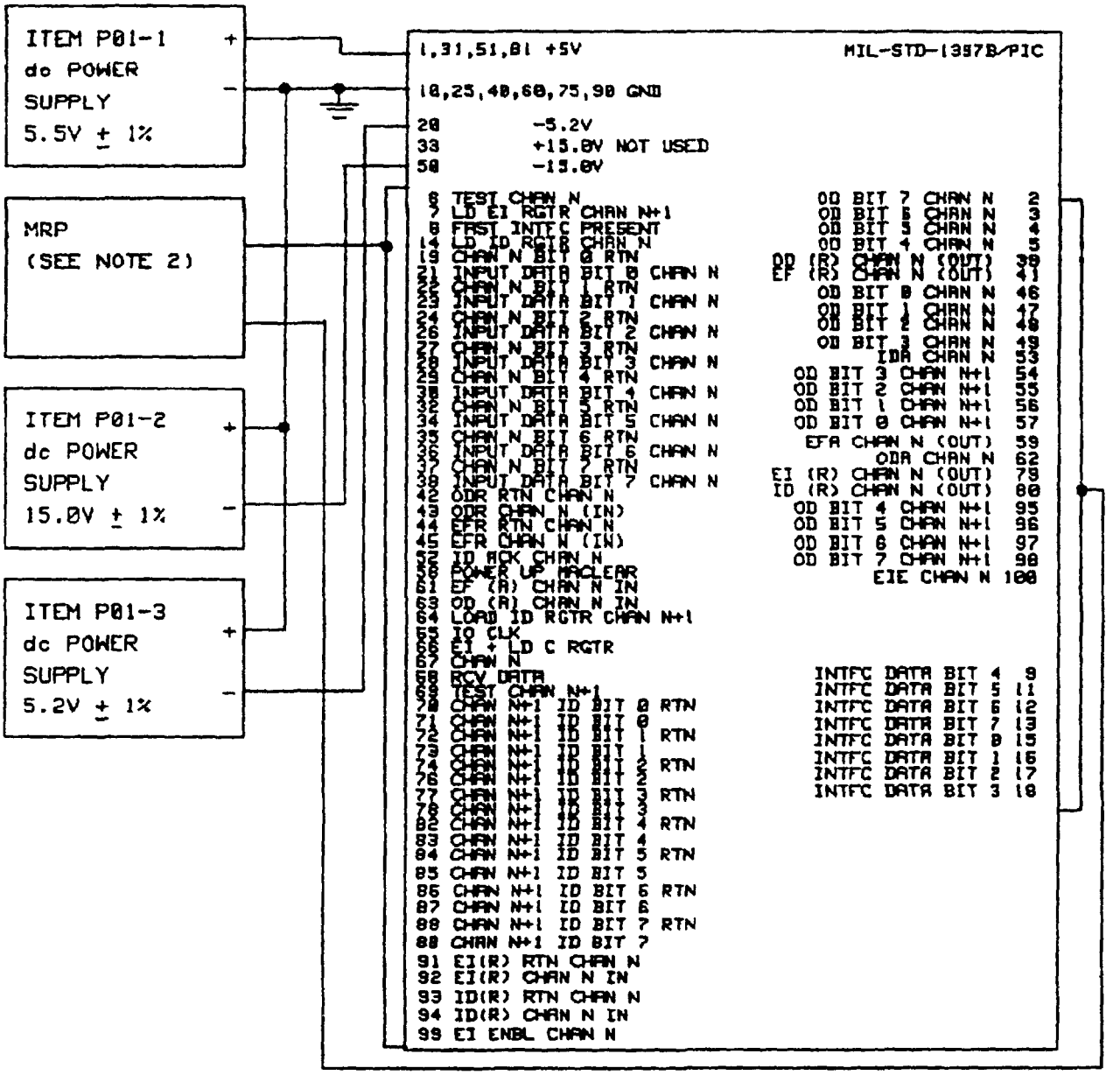


NOTE:

1. Caution: Turn the -5.2 V power supply on first and off last.

FIGURE 4. AGL vibration test circuit.

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NOTES:

1. Caution: Turn the -5.2 V power supply on first and off last.

2. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AGL life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: (Vcc) 4.5 V dc to 5.5 V dc, (V2) -5.72 V dc to -4.68 V dc, and (V3) -16.5 V dc to -13.5 V dc. Module power dissipation: 7.31 W (max); 4.86 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
TEST N	-1.5	0.4	2.4	-0.2	0.02	0.1	75
LD EI RG N+1	-1.5	0.4	2.4	-0.2	0.02	0.1	100
FAST INTFC	NA	0.4	2.4	-12.2	-2.85	NA	200
LD ID RG N	-1.5	0.4	2.4	-0.2	0.02	0.1	75
N BIT 0 RTN	NA	NA	NA	NA	NA	NA	75
N BIT 0	NA	-2.5	-0.6	NA	NA	NA	75
N BIT 1 RTN	NA	NA	NA	NA	NA	NA	75
N BIT 1	NA	-2.5	-0.6	NA	NA	NA	75
N BIT 2 RTN	NA	NA	NA	NA	NA	NA	100
N BIT 2	NA	-2.5	-0.6	NA	NA	NA	175
N BIT 3 RTN	NA	NA	NA	NA	NA	NA	75
N BIT 3	NA	-2.5	-0.6	NA	NA	NA	100
N BIT 4 RTN	NA	NA	NA	NA	NA	NA	100
N BIT 4	NA	-2.5	-0.6	NA	NA	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
N BIT 5 RTN	NA	NA	NA	NA	NA	NA	75
N BIT 5	NA	-2.5	-0.6	NA	NA	NA	100
N BIT 6 RTN	NA	NA	NA	NA	NA	NA	75
N BIT 6	NA	-2.5	-0.6	NA	NA	NA	75
N BIT 7 RTN	NA	NA	NA	NA	NA	NA	75
N BIT 7	NA	-2.5	-0.6	NA	NA	NA	75
ODR RTN N	NA	NA	NA	NA	NA	NA	75
ODR N (IN)	NA	-2.5	-0.6	NA	NA	NA	75
EFR RTN N	NA	NA	NA	NA	NA	NA	75
EFR N (IN)	NA	-2.5	-0.6	NA	NA	NA	75
ID ACK N	NA	0.4	2.4	-0.4	0.02	0.1	75
POWER UP	NA	0.4	2.4	-1.6	0.08	0.4	200
EF (A) CH N	NA	0.4	2.4	-0.4	0.02	0.1	50
OD (A) CH N	NA	0.4	2.4	-0.4	0.02	0.1	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
LD ID RG N+1	-1.5	0.4	2.4	-0.2	0.02	0.1	75
IO CLK	-1.5	0.4	2.4	-1.44	0.08	0.4	125
EI + LD C	-1.5	0.4	2.4	-0.8	0.04	0.2	100
CH N	-1.5	0.4	2.4	-0.8	0.04	0.2	100
RCV DATA	-1.5	0.4	2.4	-0.6	0.04	0.2	75
TEST N+1	-1.5	0.4	2.4	-0.2	0.02	0.1	75
N+1 ID 0 RTN	NA	NA	NA	NA	NA	NA	100
N+1 ID 0	NA	-2.5	-0.6	NA	NA	NA	100
N+1 ID 1 RTN	NA	NA	NA	NA	NA	NA	75
N+1 ID 1	NA	-2.5	-0.6	NA	NA	NA	100
N+1 ID 2 RTN	NA	NA	NA	NA	NA	NA	75
N+1 ID 2	NA	-2.5	-0.6	NA	NA	NA	75
N+1 ID 3 RTN	NA	NA	NA	NA	NA	NA	75
N+1 ID 3	NA	-2.5	-0.6	NA	NA	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
N+1 ID 4 RTN	NA	NA	NA	NA	NA	NA	75
N+1 ID 4	NA	-2.5	-0.6	NA	NA	NA	75
N+1 ID 5 RTN	NA	NA	NA	NA	NA	NA	75
N+1 ID 5	NA	-2.5	-0.6	NA	NA	NA	75
N+1 ID 6 RTN	NA	NA	NA	NA	NA	NA	75
N+1 ID 6	NA	-2.5	-0.6	NA	NA	NA	75
N+1 ID 7 RTN	NA	NA	NA	NA	NA	NA	75
N+1 ID 7	NA	-2.5	-0.6	NA	NA	NA	75
EI RTN N	NA	NA	NA	NA	NA	NA	75
EI N (IN)	NA	-2.5	-0.6	NA	NA	NA	75
ID RTN N	NA	NA	NA	NA	NA	NA	75
ID N (IN)	NA	-2.5	-0.6	NA	NA	NA	100
EI ENBL N	NA	0.4	2.4	-0.4	0.02	0.1	50
INTFC BIT 4	NA	0.4	2.4	-0.8	0.12	NA	175

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
INTFC BIT 5	NA	0.4	2.4	-0.8	0.12	NA	175
INTFC BIT 6	NA	0.4	2.4	-0.8	0.12	NA	175
INTFC BIT 7	NA	0.4	2.4	-0.8	0.12	NA	175
INTFC BIT 0	NA	0.4	2.4	-0.8	0.12	NA	175
INTFC BIT 1	NA	0.4	2.4	-0.8	0.12	NA	175
INTFC BIT 2	NA	0.4	2.4	-0.8	0.12	NA	175
INTFC BIT 3	NA	0.4	2.4	-0.8	0.12	NA	200

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
OUT 7 N	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 6 N	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 5 N	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 4 N	-0.6	-2.5	-1.5	0.5	265.0	NA
OD (R) N	2.4	0.5	-0.16	1.6	-1.3	NA
EF (R) N	2.4	0.5	-0.16	1.6	-1.3	NA
OUT 0 N	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 1 N	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 2 N	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 3 N	-0.6	-2.5	-1.5	0.5	265.0	NA
ID N OUT	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 3 N+1	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 2 N+1	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 1 N+1	-0.6	-2.5	-1.5	0.5	265.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
OUT 0 N+1	-0.6	-2.5	-1.5	0.5	265.0	NA
EFA N (OUT)	-0.6	-2.5	-1.5	0.5	265.0	NA
ODA N	-0.6	-2.5	-1.5	0.5	265.0	NA
EI (R) N	2.4	0.5	NA	1.6	-5.24	NA
ID (R) N	2.4	0.5	NA	1.6	-5.24	NA
OUT 4 N+1	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 5 N+1	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 6 N+1	-0.6	-2.5	-1.5	0.5	265.0	NA
OUT 7 N+1	-0.6	-2.5	-1.5	0.5	265.0	NA
EIE N	-0.6	-2.5	-1.5	0.5	265.0	NA
INTFC BIT 4	2.4	0.5	-0.16	1.6	-30.0	175
INTFC BIT 5	2.4	0.5	-0.16	1.6	-30.0	175
INTFC BIT 6	2.4	0.5	-0.16	1.6	-30.0	175
INTFC BIT 7	2.4	0.5	-0.16	1.6	-30.0	175

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
INTFC BIT 0	2.4	0.5	-0.16	1.6	-30.0	175
INTFC BIT 1	2.4	0.5	-0.16	1.6	-30.0	175
INTFC BIT 2	2.4	0.5	-0.16	1.6	-30.0	175
INTFC BIT 3	2.4	0.5	-0.16	1.6	-30.0	200

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1 1/	tsu	INTFC BIT 0 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 1 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 2 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 3 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 4 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 5 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 6 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 7 to IO CLK	15	15	15	-	-	-	nS	12
2 2/	tsu	INTFC BIT 0 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 1 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 2 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 3 to IO CLK	15	15	15	-	-	-	nS	12

1/ INTFC BIT 0 - INTFC BIT 7 is channel N.

2/ INTFC BIT 0 - INTFC BIT 7 is channel N+1.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2 2/	tsu	INTFC BIT 4 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 5 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 6 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 7 to IO CLK	15	15	15	-	-	-	nS	12
3 1/	th	IO CLK to INTFC BIT 0	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 1	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 2	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 3	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 4	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 5	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 6	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 7	10	10	10	-	-	-	nS	12
4 2/	th	IO CLK to INTFC BIT 0	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 1	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 2	10	10	10	-	-	-	nS	12

1/ INTFC BIT 0 - INTFC BIT 7 is channel N.

2/ INTFC BIT 0 - INTFC BIT 7 is channel N+1.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4 2/	th	IO CLK to INTFC BIT 3	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 4	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 5	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 6	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 7	10	10	10	-	-	-	nS	12
5	tpd	IO CLK to OUT 0 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 1 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 2 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 3 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 4 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 5 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 6 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 7 N	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 0 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 1 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 2 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 3 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 4 N+1	-	-	-	470	470	470	nS	12

2/ INTFC BIT 0 - INTFC BIT 7 is channel N+1.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	IO CLK to OUT 5 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 6 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 7 N+1	-	-	-	470	470	470	nS	12
7	tpd	N BIT 0 to INTFC BIT 0	-	-	-	115	115	115	nS	12
7	tpd	N BIT 1 to INTFC BIT 1	-	-	-	115	115	115	nS	12
7	tpd	N BIT 2 to INTFC BIT 2	-	-	-	115	115	115	nS	12
7	tpd	N BIT 3 to INTFC BIT 3	-	-	-	115	115	115	nS	12
7	tpd	N BIT 4 to INTFC BIT 4	-	-	-	115	115	115	nS	12
7	tpd	N BIT 5 to INTFC BIT 5	-	-	-	115	115	115	nS	12
7	tpd	N BIT 6 to INTFC BIT 6	-	-	-	115	115	115	nS	12
7	tpd	N BIT 7 to INTFC BIT 7	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 0 to INTFC BIT 0	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 1 to INTFC BIT 1	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 2 to INTFC BIT 2	-	-	-	115	115	115	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	N+1 ID 3 to INTFC BIT 3	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 4 to INTFC BIT 4	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 5 to INTFC BIT 5	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 6 to INTFC BIT 6	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 7 to INTFC BIT 7	-	-	-	115	115	115	nS	12
9 1/	tpd	EI + LD C & CH N to INTFC BIT 0	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 1	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 2	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 3	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 4	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 5	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 6	-	-	-	65	65	65	nS	12

1/ INTFC BIT 0 - INTFC BIT 7 is channel N.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9 1/	tpd	EI + LD C & CH N to INTFC BIT 7	-	-	-	65	65	65	nS	12
10 2/	tpd	EI + LD C & CH N to INTFC BIT 0	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 1	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 2	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 3	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 4	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 5	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 6	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 7	-	-	-	65	65	65	nS	12

1/ INTFC BIT 0 - INTFC BIT 7 is channel N.

2/ INTFC BIT 0 - INTFC BIT 7 is channel N+1.

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6.4 Module function. The parallel interface modules (type A, B, and C) contain input and output data circuits, request receivers and read/write control circuits. Each of the modules handle 8 bits of data for channel N and channel N+1. Functionally, MIL-STD-1397 type A, B, and C interfaces operate identically; the differences being in the receiver and driver circuits for the particular interface. Input data receivers receive data or interrupt codes and transfer this information to an External Interrupt/Input Data (EI/ID) register. Loading of these registers is controlled by the read/write control circuits. Output circuits consist of an output data register and output data driver. The output data driver will transmit data and acknowledge signals at the voltage level corresponding to the selected interface. The request receivers receive the channel requests (External Function Request (EFR), External Interrupt Request (EIR), Output Data Request (ODR), Input Data Request (IDR)) and transfer them to the scan, request and timing control circuits on the parallel channel adapter module.

6.4.1 Input data receivers. Inputs to the input data receivers are information outputs from a peripheral device characterized by the following levels:

(a) Nominal values of 0 volts and -15 volts to represent a "1" and a "0" respectively, with a switching threshold of -6 $\pm$ 1.5 volts (type A).

(b) Nominal values of 0 volts and -3 volts to represent a "1" and a "0" respectively, with a switching threshold of -1.5 $\pm$ 0.4 volts (type B).

(c) Nominal values of 0 volts and +3.5 volts to represent a "1" and a "0" respectively, with a switching threshold of +0.8 and +2.2 volts (type C).

Line receivers and passive resistive/capacitive networks are employed with proper biasing levels to ensure compliance with the interface levels described above. Outputs from the receivers are applied to input data registers and external interrupt registers. The test channel N and N+1 signals are also applied to the module at the receivers from the Input/Output (I/O) parallel channel control module for channel testing.

6.4.2 Input data register. The 8-bit input data register holds the input data byte. The register is controlled by the parallel channel control module (ld ID rgtr chan N/N+1).

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The register is controlled by rd ID rgtr chan N/N+1 also from the parallel channel control module. Outputs from the input data register are placed on the interface data bus and subsequently sent to the parallel channel control module.

6.4.3 External interrupt register. The 8-bit external interrupt register holds the interrupt word read from the peripheral device via the input data receivers. The register is enabled by the ld EI rgtr chan N/N+1 signal from the read/write control circuits. The load signal is a gated EIR from the request receivers. The register is controlled by the rd EI rgtr chan N/N+1 signal from the read/write control circuits. Outputs from the external interrupt register are placed on the interface data bus and sent to the parallel channel control module.

6.4.4 Output data register. The 8-bit output data register holds the data that is destined to be transmitted to the peripheral device. Levels are characterized by the values listed in 6.4.1. The output data register is clocked by the IO clock and enabled by the ld c regtr chan N/N+1 signal from the read/write control circuits. Test signals are also applied from the parallel I/O channel control module to allow module test.

6.4.5 Output data drivers. The output data drivers convert the Transistor Transistor Logic (TTL) output data into the required level for the interface type and drive the output data lines.

6.4.6 Request receivers. The request receivers convert the IDR, EIR, ODR, and EFR from the interface level (described in 6.4.1) to TTL level. Outputs from the request receiver are applied to the parallel channel control module. The IDR and EIR signals are gated with the test chan N/N+1 signal for module testing.

6.4.7 Read/write circuits. The read/write control circuits perform the various load, test and control functions for the MIL-STD-1397 types A, B, and C modules. A pair of two-line and four-line decoders accomplish the task of differentiating between elements of channel N and channel N+1 in addition to providing the read signals for the input data and external input registers. Test signals for channel N and channel N+1 are gated with EIR and IDR to provide module testing for both channels.

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6.4.8 Description of I/O pins.

Signal name	Pin number	I/O	Description
TEST N	6	I	TEST CHAN N
LD EI RG N+1	7	I	LD EI RGTR CHAN N+1
FAST INTFC	8	I	FAST INTFC PRESENT
LD ID RG N	14	I	LD ID RGTR CHAN N
N BIT 0 RTN	19	I	CHAN N BIT 0 RTN
N BIT 0	21	I	INPUT DATA BIT 0 CHAN N
N BIT 1 RTN	22	I	CHAN N BIT 1 RTN
N BIT 1	23	I	INPUT DATA BIT 1 CHAN N
N BIT 2 RTN	24	I	CHAN N BIT 2 RTN
N BIT 2	26	I	INPUT DATA BIT 2 CHAN N
N BIT 3 RTN	27	I	CHAN N BIT 3 RTN
N BIT 3	28	I	INPUT DATA BIT 3 CHAN N
N BIT 4 RTN	29	I	CHAN N BIT 4 RTN
N BIT 4	30	I	INPUT DATA BIT 4 CHAN N
N BIT 5 RTN	32	I	CHAN N BIT 5 RTN
N BIT 5	34	I	INPUT DATA BIT 5 CHAN N
N BIT 6 RTN	35	I	CHAN N BIT 6 RTN
N BIT 6	36	I	INPUT DATA BIT 6 CHAN N
N BIT 7 RTN	37	I	CHAN N BIT 7 RTN
N BIT 7	38	I	INPUT DATA BIT 7 CHAN N

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ODR RTN N	42	I	ODR RTN CHAN N
ODR N (IN)	43	I	ODR CHAN N (IN)
EFR RTN N	44	I	EFR RTN CHAN N
EFR N (IN)	45	I	EFR CHAN N (IN)
ID ACK N	52	I	ID ACK CHAN N
POWER UP	58	I	POWER UP MACLEAR
EF (A) CH N	61	I	EF (A) CHAN N IN
OD (A) CH N	63	I	OD (A) CHAN N IN
LD ID RG N+1	64	I	LOAD ID RGTR CHAN N+1
IO CLK	65	I	IO CLK
EI + LD C	66	I	EI + LD C RGTR
CH N	67	I	CHAN N
RCV DATA	68	I	RCV DATA
TEST N+1	69	I	TEST CHAN N+1
N+1 ID 0 RTN	70	I	CHAN N+1 ID BIT 0 RTN
N+1 ID 0	71	I	CHAN N+1 ID BIT 0
N+1 ID 1 RTN	72	I	CHAN N+1 ID BIT 1 RTN
N+1 ID 1	73	I	CHAN N+1 ID BIT 1
N+1 ID 2 RTN	74	I	CHAN N+1 ID BIT 2 RTN
N+1 ID 2	76	I	CHAN N+1 ID BIT 2

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
N+1 ID 3 RTN	77	I	CHAN N+1 ID BIT 3 RTN
N+1 ID 3	78	I	CHAN N+1 ID BIT 3
N+1 ID 4 RTN	82	I	CHAN N+1 ID BIT 4 RTN
N+1 ID 4	83	I	CHAN N+1 ID BIT 4
N+1 ID 5 RTN	84	I	CHAN N+1 ID BIT 5 RTN
N+1 ID 5	85	I	CHAN N+1 ID BIT 5
N+1 ID 6 RTN	86	I	CHAN N+1 ID BIT 6 RTN
N+1 ID 6	87	I	CHAN N+1 ID BIT 6
N+1 ID 7 RTN	88	I	CHAN N+1 ID BIT 7 RTN
N+1 ID 7	89	I	CHAN N+1 ID BIT 7
EI RTN N	91	I	EI(R) RTN CHAN N
EI N (IN)	92	I	EI(R) CHAN N IN
ID RTN N	93	I	ID(R) RTN CHAN N
ID N (IN)	94	I	ID(R) CHAN N IN
EI ENBL N	99	I	EI ENBL CHAN N
OUT 7 N	2	O	OD BIT 7 CHAN N
OUT 6 N	3	O	OD BIT 6 CHAN N
OUT 5 N	4	O	OD BIT 5 CHAN N
OUT 4 N	5	O	OD BIT 4 CHAN N
OD (R) N	39	O	OD (R) CHAN N (OUT)

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
EF (R) N	41	O	EF (R) CHAN N (OUT)
OUT 0 N	46	O	OD BIT 0 CHAN N
OUT 1 N	47	O	OD BIT 1 CHAN N
OUT 2 N	48	O	OD BIT 2 CHAN N
OUT 3 N	49	O	OD BIT 3 CHAN N
ID N OUT	53	O	IDA CHAN N
OUT 3 N+1	54	O	OD BIT 3 CHAN N+1
OUT 2 N+1	55	O	OD BIT 2 CHAN N+1
OUT 1 N+1	56	O	OD BIT 1 CHAN N+1
OUT 0 N+1	57	O	OD BIT 0 CHAN N+1
EFA N (OUT)	59	O	EFA CHAN N (OUT)
ODA N	62	O	ODA CHAN N
EI (R) N	79	O	EI (R) CHAN N (OUT)
ID (R) N	80	O	ID (R) CHAN N (OUT)
OUT 4 N+1	95	O	OD BIT 4 CHAN N+1
OUT 5 N+1	96	O	OD BIT 5 CHAN N+1
OUT 6 N+1	97	O	OD BIT 6 CHAN N+1
OUT 7 N+1	98	O	OD BIT 7 CHAN N+1
EIE N	100	O	EIE CHAN N
INTFC BIT 4	9	I/O	INTFC DATA BIT 4

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6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
INTFC BIT 5	11	I/O	INTFC DATA BIT 5
INTFC BIT 6	12	I/O	INTFC DATA BIT 6
INTFC BIT 7	13	I/O	INTFC DATA BIT 7
INTFC BIT 0	15	I/O	INTFC DATA BIT 0
INTFC BIT 1	16	I/O	INTFC DATA BIT 1
INTFC BIT 2	17	I/O	INTFC DATA BIT 2
INTFC BIT 3	18	I/O	INTFC DATA BIT 3

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N727)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C,33	Frame	10	INF	MOhm	1-8
A2	Isolation, unused to used pins	33	C	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Isolation measurement	NA	± 500 KOhm	A1,A2

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal, and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

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NOTES: - Continued.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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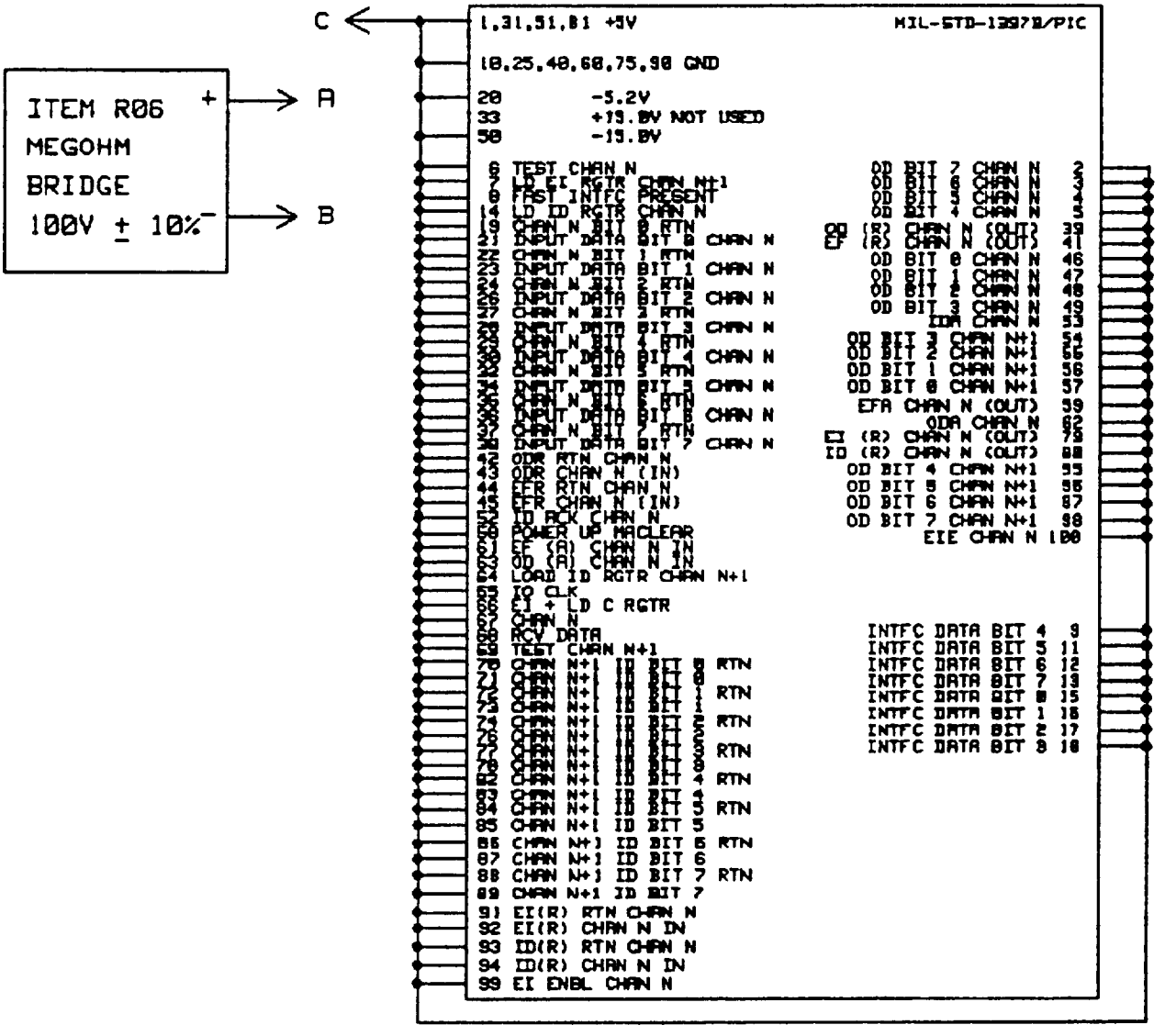


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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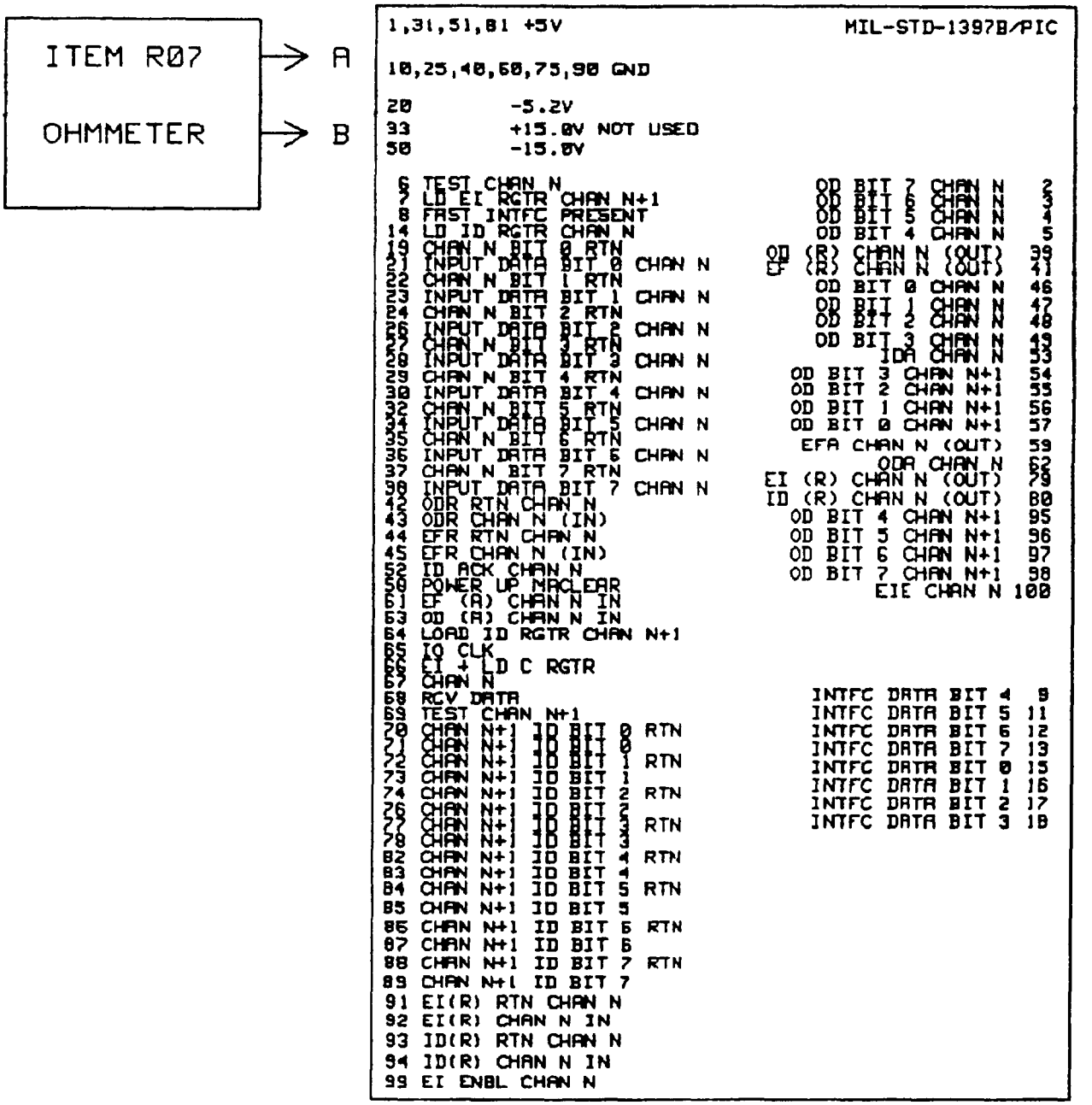


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6
C	V2	7.0	10,25, 40,60, 75,90	20	Module shall pass all remaining tests.	1-6
C	V3	17.5	10,25, 40,60, 75,90	50	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
(V2)	-7.0 V	+70 mV,-0 mV	C
(V3)	-17.5 V	+175 mV,-0 mV	C
Power supply current limit, (Vcc)	1.44 A	NA	C
(V2)	-0.13 A	NA	C
(V3)	-0.28 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

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NOTES: - Continued.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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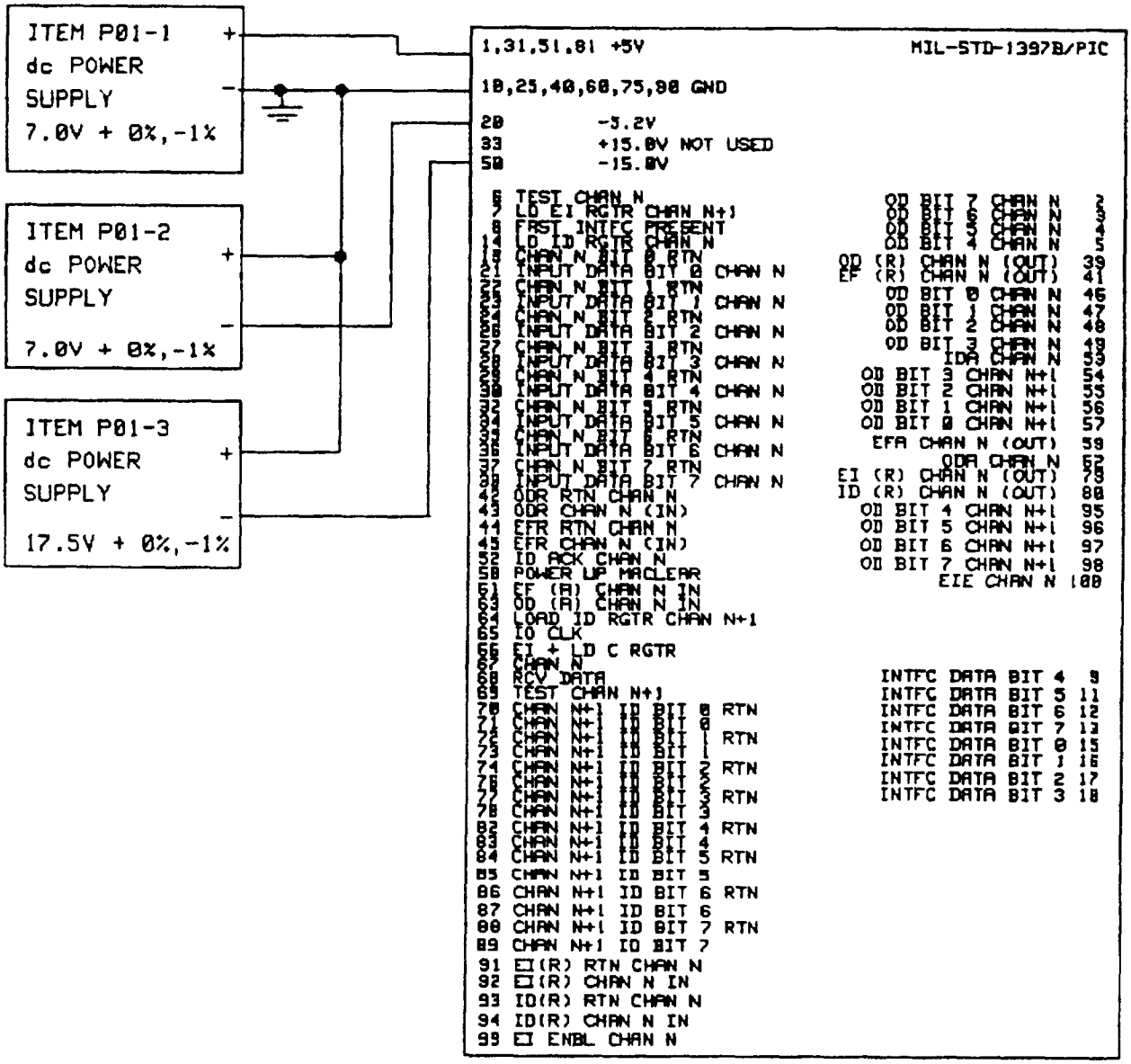


FIGURE 8. Power supply stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	TEST N	6	5.5	0.0	0.1	mA	1-3
D2	LD EI RG N+1	7					
D3	LD ID RG N	14					
D4	ID ACK N	52	5.5	0.0	0.1	mA	1-3
D5	POWER UP	58	5.5	0.0	0.4	mA	1-3
D6	EF (A) CH N	61	5.5	0.0	0.1	mA	1-3
D7	OD (A) CH N	63					
D8	LD ID RG N+1	64	5.5	0.0	0.1	mA	1-3
D9	IO CLK	65	5.5	0.0	0.4	mA	1-3
D10	EI + LD C	66	5.5	0.0	0.2	mA	1-3
D11	CH N	67					
D12	RCV DATA	68	5.5	0.0	0.2	mA	1-3
D13	TEST N+1	69	5.5	0.0	0.1	mA	1-3
D14	EI ENBL N	99	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D14
(V2)	-5.72 V	± 55 mV	D1-D14
(V3)	-16.5 V	± 165 mV	D1-D14
Power supply current limit, (Vcc)	1.18 A	NA	D1-D14
(V2)	-0.10 A	NA	D1-D14
(V3)	-0.23 A	NA	D1-D14
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D14
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D14
low, (VL)	0.0 V	± 100 mV	D1-D14
Input current measurement	NA	± 5 uA	D1-D4
	NA	± 20 uA	D5
	NA	± 5 uA	D6-D8
	NA	± 20 uA	D9
	NA	± 10 uA	D10-D12
	NA	± 5 uA	D13,D14

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	OUT 7 N	2	0.0	5.0	265.0	mA	1-4
E2	OUT 6 N	3					
E3	OUT 5 N	4					
E4	OUT 4 N	5	0.0	5.0	265.0	mA	1-4
E5	INTFC BIT 4	9	0.0	-220.0	-30.0	mA	1-4
E6	INTFC BIT 4	9					
E7	INTFC BIT 4	9					
E8	INTFC BIT 4	9					
E9	INTFC BIT 5	11					
E10	INTFC BIT 5	11					
E11	INTFC BIT 5	11					
E12	INTFC BIT 5	11					
E13	INTFC BIT 6	12					
E14	INTFC BIT 6	12					
E15	INTFC BIT 6	12					
E16	INTFC BIT 6	12					
E17	INTFC BIT 7	13					
E18	INTFC BIT 7	13					
E19	INTFC BIT 7	13					
E20	INTFC BIT 7	13	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	INTFC BIT 0	15	0.0	-220.0	-30.0	mA	1-4
E22	INTFC BIT 0	15					
E23	INTFC BIT 0	15					
E24	INTFC BIT 0	15					
E25	INTFC BIT 1	16					
E26	INTFC BIT 1	16					
E27	INTFC BIT 1	16					
E28	INTFC BIT 1	16					
E29	INTFC BIT 2	17					
E30	INTFC BIT 2	17					
E31	INTFC BIT 2	17					
E32	INTFC BIT 2	17					
E33	INTFC BIT 3	18					
E34	INTFC BIT 3	18					
E35	INTFC BIT 3	18					
E36	INTFC BIT 3	18	0.0	-220.0	-30.0	mA	1-4
E37	OD (R) N	39	0.0	-3.1	-1.3	mA	1-4
E38	EF (R) N	41	0.0	-3.1	-1.3	mA	1-4
E39	OUT 0 N	46	0.0	5.0	265.0	mA	1-4
E40	OUT 1 N	47	0.0	5.0	265.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	OUT 2 N	48	0.0	5.0	265.0	mA	1-4
E42	OUT 3 N	49					
E43	ID N OUT	53					
E44	OUT 3 N+1	54					
E45	OUT 2 N+1	55					
E46	OUT 1 N+1	56					
E47	OUT 0 N+1	57					
E48	EFA N (OUT)	59					
E49	ODA N	62	0.0	5.0	265.0	mA	1-4
E50	EI (R) N	79	0.0	-6.04	-5.24	mA	1-4
E51	ID (R) N	80	0.0	-6.04	-5.24	mA	1-4
E52	OUT 4 N+1	95	0.0	5.0	265.0	mA	1-4
E53	OUT 5 N+1	96					
E54	OUT 6 N+1	97					
E55	OUT 7 N+1	98					
E56	EIE N	100	0.0	5.0	265.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E56
(V2)	-5.72 V	± 55 mV	E1-E56
(V3)	-16.5 V	± 165 mV	E1-E56
Power supply current limit, (Vcc)	1.18 A	NA	E1-E56
(V2)	-0.10 A	NA	E1-E56
(V3)	-0.23 A	NA	E1-E56
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E56
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E56
low, (VL)	0.0 V	± 100 mV	E1-E56
Output current measurement	NA	± 13.00 mA	E1-E4
	NA	± 9.50 mA	E5-E36
	NA	± 0.09 mA	E37,E38
	NA	± 13.00 mA	E39-E49
	NA	± 0.04 mA	E50,E51
	NA	± 13.00 mA	E52-E56

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	TEST N	6	2.4	-0.2	0.02	mA	1-3
F2	LD EI RG N+1	7	2.4	-0.2	0.02	mA	1-3
F3	FAST INTFC	8	2.4	-12.2	-2.85	mA	1-3
F4	INTFC BIT 4	9	2.4	-0.8	0.12	mA	1-3
F5	INTFC BIT 5	11					
F6	INTFC BIT 6	12					
F7	INTFC BIT 7	13	2.4	-0.8	0.12	mA	1-3
F8	LD ID RG N	14	2.4	-0.2	0.02	mA	1-3
F9	INTFC BIT 0	15	2.4	-0.8	0.12	mA	1-3
F10	INTFC BIT 1	16					
F11	INTFC BIT 2	17					
F12	INTFC BIT 3	18	2.4	-0.8	0.12	mA	1-3
F13	ID ACK N	52	2.4	-0.4	0.02	mA	1-3
F14	POWER UP	58	2.4	-1.6	0.08	mA	1-3
F15	EF (A) CH N	61	2.4	-0.4	0.02	mA	1-3
F16	OD (A) CH N	63	2.4	-0.4	0.02	mA	1-3
F17	LD ID RG N+1	64	2.4	-0.2	0.02	mA	1-3
F18	IO CLK	65	2.4	-1.44	0.08	mA	1-3
F19	EI + LD C	66	2.4	-0.8	0.04	mA	1-3
F20	CH N	67	2.4	-0.8	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F21	RCV DATA	68	2.4	-0.6	0.04	mA	1-3
F22	TEST N+1	69	2.4	-0.2	0.02	mA	1-3
F23	EI ENBL N	99	2.4	-0.4	0.02	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F23
(V2)	-5.72 V	± 55 mV	F1-F23
(V3)	-16.5 V	± 165 mV	F1-F23
Power supply current limit, (Vcc)	1.18 A	NA	F1-F23
(V2)	-0.10 A	NA	F1-F23
(V3)	-0.23 A	NA	F1-F23
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F23
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F23
low, (VL)	0.0 V	± 100 mV	F1-F23
Input current measurement	NA	± 11 uA	F1,F2
	NA	± 470 uA	F3
	NA	± 46 uA	F4-F7
	NA	± 11 uA	F8
	NA	± 46 uA	F9-F12
	NA	± 21 uA	F13
	NA	± 84 uA	F14
	NA	± 21 uA	F15,F16
	NA	± 11 uA	F17
	NA	± 76 uA	F18
	NA	± 42 uA	F19,F20
	NA	± 32 uA	F21
	NA	± 11 ua	F22
	NA	± 21 uA	F23

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	TEST N	6	0.4	-0.2	0.02	mA	1-3
G2	LD EI RG N+1	7	0.4	-0.2	0.02	mA	1-3
G3	FAST INTFC	8	0.4	-12.2	-2.85	mA	1-3
G4	INTFC BIT 4	9	0.4	-0.8	0.12	mA	1-3
G5	INTFC BIT 5	11					
G6	INTFC BIT 6	12					
G7	INTFC BIT 7	13	0.4	-0.8	0.12	mA	1-3
G8	LD ID RG N	14	0.4	-0.2	0.02	mA	1-3
G9	INTFC BIT 0	15	0.4	-0.8	0.12	mA	1-3
G10	INTFC BIT 1	16					
G11	INTFC BIT 2	17					
G12	INTFC BIT 3	18	0.4	-0.8	0.12	mA	1-3
G13	ID ACK N	52	0.4	-0.4	0.02	mA	1-3
G14	POWER UP	58	0.4	-1.6	0.08	mA	1-3
G15	EF (A) CH N	61	0.4	-0.4	0.02	mA	1-3
G16	OD (A) CH N	63	0.4	-0.4	0.02	mA	1-3
G17	LD ID RG N+1	64	0.4	-0.2	0.02	mA	1-3
G18	IO CLK	65	0.4	-1.44	0.08	mA	1-3
G19	EI + LD C	66	0.4	-0.8	0.04	mA	1-3
G20	CH N	67	0.4	-0.8	0.04	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	RCV DATA	68	0.4	-0.6	0.04	mA	1-3
G22	TEST N+1	69	0.4	-0.2	0.02	mA	1-3
G23	EI ENBL N	99	0.4	-0.4	0.02	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G23
(V2)	-5.72 V	± 55 mV	G1-G23
(V3)	-16.5 V	± 165 mV	G1-G23
Power supply current limit, (Vcc)	1.18 A	NA	G1-G23
(V2)	-0.10 A	NA	G1-G23
(V3)	-0.23 A	NA	G1-G23
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G23
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G23
low, (VL)	0.0 V	± 100 mV	G1-G23
Input current measurement	NA	± 11 uA	G1,G2
	NA	± 470 uA	G3
	NA	± 46 uA	G4-G7

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 11 uA	G8
	NA	± 46 uA	G9-G12
	NA	± 21 uA	G13
	NA	± 84 uA	G14
	NA	± 21 uA	G15,G16
	NA	± 11 uA	G17
	NA	± 76 uA	G18
	NA	± 42 uA	G19,G20
	NA	± 32 uA	G21
	NA	± 11 uA	G22
	NA	± 21 uA	G23

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	OUT 7 N	2	-1.5	-0.6	0.0	V	1-4
H2	OUT 6 N	3					
H3	OUT 5 N	4					
H4	OUT 4 N	5	-1.5	-0.6	0.0	V	1-4
H5	INTFC BIT 4	9	-0.16	2.4	4.5	V	1-4
H6	INTFC BIT 4	9					
H7	INTFC BIT 4	9					
H8	INTFC BIT 4	9					
H9	INTFC BIT 5	11					
H10	INTFC BIT 5	11					
H11	INTFC BIT 5	11					
H12	INTFC BIT 5	11					
H13	INTFC BIT 6	12					
H14	INTFC BIT 6	12					
H15	INTFC BIT 6	12					
H16	INTFC BIT 6	12					
H17	INTFC BIT 7	13					
H18	INTFC BIT 7	13					
H19	INTFC BIT 7	13					
H20	INTFC BIT 7	13	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	INTFC BIT 0	15	-0.16	2.4	4.5	V	1-4
H22	INTFC BIT 0	15					
H23	INTFC BIT 0	15					
H24	INTFC BIT 0	15					
H25	INTFC BIT 1	16					
H26	INTFC BIT 1	16					
H27	INTFC BIT 1	16					
H28	INTFC BIT 1	16					
H29	INTFC BIT 2	17					
H30	INTFC BIT 2	17					
H31	INTFC BIT 2	17					
H32	INTFC BIT 2	17					
H33	INTFC BIT 3	18					
H34	INTFC BIT 3	18					
H35	INTFC BIT 3	18					
H36	INTFC BIT 3	18					
H37	OD (R) N	39					
H38	EF (R) N	41	-0.16	2.4	4.5	V	1-4
H39	OUT 0 N	46	-1.5	-0.6	0.0	V	1-4
H40	OUT 1 N	47	-1.5	-0.6	0.0	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H41	OUT 2 N	48	-1.5	-0.6	0.0	V	1-4
H42	OUT 3 N	49					
H43	ID N OUT	53					
H44	OUT 3 N+1	54					
H45	OUT 2 N+1	55					
H46	OUT 1 N+1	56					
H47	OUT 0 N+1	57					
H48	EFA N (OUT)	59					
H49	ODA N	62	-1.5	-0.6	0.0	V	1-4
H50	OUT 4 N+1	95	-1.5	-0.6	0.0	V	1-4
H51	OUT 5 N+1	96					
H52	OUT 6 N+1	97					
H53	OUT 7 N+1	98					
H54	EIE N	100	-1.5	-0.6	0.0	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc) (V2) (V3)	4.5 V -4.7 V -13.5 V	± 45 mV ± 45 mV ± 135 mV	H1-H54 H1-H54 H1-H54
Power supply current limit, (Vcc) (V2) (V3)	1.18 A -0.10 A -0.23 A	NA NA NA	H1-H54 H1-H54 H1-H54
Forcing current, (Ioh)	-1.5 mA -0.16 mA -1.5 mA -1.5 mA	± 150 uA ± 16 uA ± 150 uA ± 150 uA	H1-H4 H5-H38 H39-H49 H50-H54
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H54 H1-H54
Output voltage measurement	NA NA NA NA	± 30 mV ± 105 mV ± 30 mV ± 30 mV	H1-H4 H5-H38 H39-H49 H50-H54

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

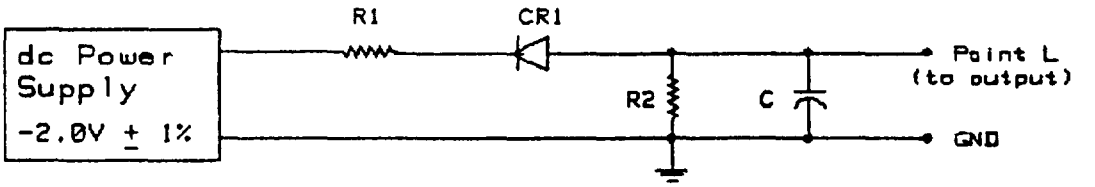
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

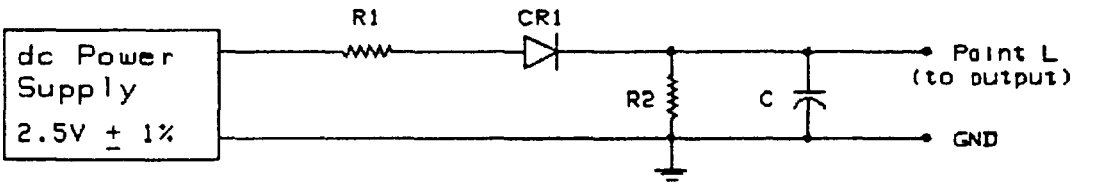
3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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Load circuit A



Load circuit B

NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. The appropriate logic load circuit for each output pin is specified in the table below.

4. The value of R1 and R2 for each output load shall be as specified in the table below.

Pin number	R1 1/ (ohms)	R2 1/ (ohms)	Logic load circuit
2,3,4,5,46,47,48, 49,53,54,55,56,57, 59,62,95,96,97,98, 100	1000	9100	A
9,11,12,13,15,16, 17,18,39,41,79,80	750	16000	B

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	OUT 7 N	2	0.5	-4.5	-2.5	V	1-4
J2	OUT 6 N	3					
J3	OUT 5 N	4					
J4	OUT 4 N	5	0.5	-4.5	-2.5	V	1-4
J5	INTFC BIT 4	9	1.6	0.0	0.5	V	1-4
J6	INTFC BIT 4	9					
J7	INTFC BIT 4	9					
J8	INTFC BIT 4	9					
J9	INTFC BIT 5	11					
J10	INTFC BIT 5	11					
J11	INTFC BIT 5	11					
J12	INTFC BIT 5	11					
J13	INTFC BIT 6	12					
J14	INTFC BIT 6	12					
J15	INTFC BIT 6	12					
J16	INTFC BIT 6	12					
J17	INTFC BIT 7	13					
J18	INTFC BIT 7	13					
J19	INTFC BIT 7	13					
J20	INTFC BIT 7	13	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	INTFC BIT 0	15	1.6	0.0	0.5	V	1-4
J22	INTFC BIT 0	15					
J23	INTFC BIT 0	15					
J24	INTFC BIT 0	15					
J25	INTFC BIT 1	16					
J26	INTFC BIT 1	16					
J27	INTFC BIT 1	16					
J28	INTFC BIT 1	16					
J29	INTFC BIT 2	17					
J30	INTFC BIT 2	17					
J31	INTFC BIT 2	17					
J32	INTFC BIT 2	17					
J33	INTFC BIT 3	18					
J34	INTFC BIT 3	18					
J35	INTFC BIT 3	18					
J36	INTFC BIT 3	18					
J37	OD (R) N	39					
J38	EF (R) N	41	1.6	0.0	0.5	V	1-4
J39	OUT 0 N	46	0.5	-4.5	-2.5	V	1-4
J40	OUT 1 N	47	0.5	-4.5	-2.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	OUT 2 N	48	0.5	-4.5	-2.5	V	1-4
J42	OUT 3 N	49					
J43	ID N OUT	53					
J44	OUT 3 N+1	54					
J45	OUT 2 N+1	55					
J46	OUT 1 N+1	56					
J47	OUT 0 N+1	57					
J48	EFA N (OUT)	59					
J49	ODA N	62	0.5	-4.5	-2.5	V	1-4
J50	EI (R) N	79	1.6	0.0	0.5	V	1-4
J51	ID (R) N	80	1.6	0.0	0.5	V	1-4
J52	OUT 4 N+1	95	0.5	-4.5	-2.5	V	1-4
J53	OUT 5 N+1	96					
J54	OUT 6 N+1	97					
J55	OUT 7 N+1	98					
J56	EIE N	100	0.5	-4.5	-2.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J56
(V2)	-4.7 V	± 45 mV	J1-J56
(V3)	-13.5 V	± 135 mV	J1-J56
Power supply current limit, (Vcc)	1.18 A	NA	J1-J56
(V2)	-0.10 A	NA	J1-J56
(V3)	-0.23 A	NA	J1-J56
Forcing current, (Iol)	0.5 mA	± 50 uA	J1-J4
	1.6 mA	± 160 uA	J5-J38
	0.5 mA	± 50 uA	J39-J49
	1.6 mA	± 160 uA	J50,J51
	0.5 mA	± 50 uA	J52-J56
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J56
low, (VL)	0.0 V	± 100 mV	J1-J56
Output voltage measurement	NA	± 100 mV	J1-J4
	NA	± 25 mV	J5-J38
	NA	± 100 mV	J39-J49
	NA	± 25 mV	J50,J51
	NA	± 100 mV	J52-J56

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	TEST N	6	-18.0	-1.5	0.0	V	1-3
L2	LD EI RG N+1	7					
L3	LD ID RG N	14					
L4	LD ID RG N+1	64					
L5	IO CLK	65					
L6	EI + LD C	66					
L7	CH N	67					
L8	RCV DATA	68					
L9	TEST N+1	69	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L9
(V2)	Off	NA	L1-L9
(V3)	Off	NA	L1-L9
Power supply current limit, (Vcc)	NA	NA	L1-L9
(V2)	NA	NA	L1-L9
(V3)	NA	NA	L1-L9
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L9

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L9 L1-L9
Input voltage measurement	NA	± 75 mV	L1-L9

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC, Vcc	1	4.4	8.2	uF	1-4
M2	PSFC, Vcc	51	3.8	7.0	uF	1-4
M3	PSFC, V2	20	11.0	20.2	uF	1-4
M4	PSFC, V3	50	1.4	2.6	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1-M4
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1-M4
Capacitance measurement	NA NA NA NA	± 190 nF ± 160 nF ± 460 nF ± 60 nF	M1 M2 M3 M4

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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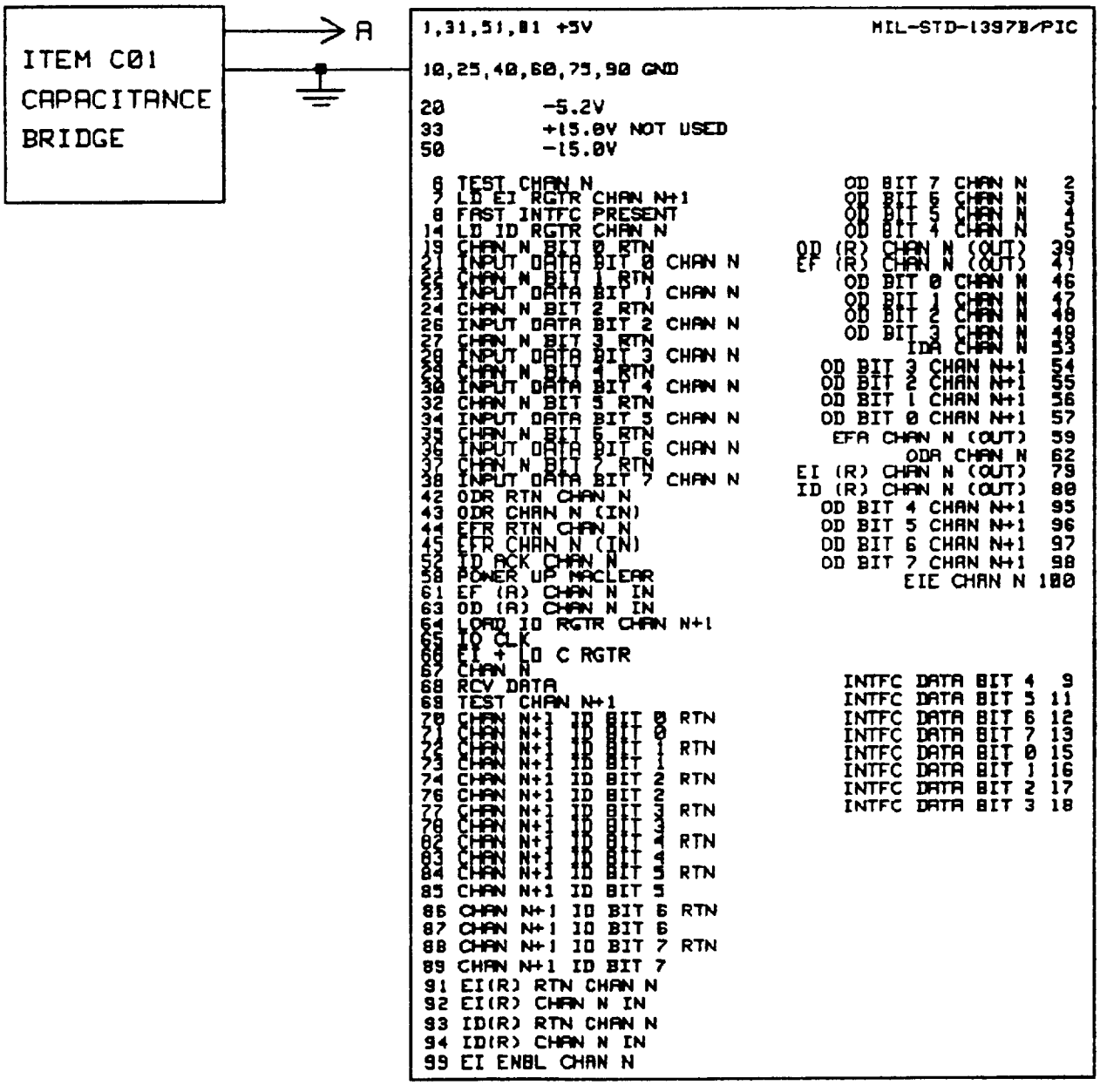


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	Icc, Vcc	1,31, 51,81	330.0	790.0	mA	1-3
N2	Icc, V2	20	-67.0	-23.75	mA	1-3
N3	Icc, V3	50	-156.0	-60.0	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1-N3
(V2)	-5.72 V	± 55 mV	N1-N3
(V3)	-16.5 V	± 165 mV	N1-N3
Power supply current limit, (Vcc)	1.18 A	NA	N1-N3
(V2)	-0.10 A	NA	N1-N3
(V3)	-0.23 A	NA	N1-N3
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1-N3
low, (VL)	0.0 V	± 100 mV	N1-N3
Power supply current measurement	NA	± 23.0 mA	N1
	NA	± 2.16 mA	N2
	NA	± 4.80 mA	N3

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (1000 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (1000 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	P1
(V2)	-5.2 V	± 55 mV	P1,P2
(V3)	-15.0 V	± 150 mV	P1,P2
(Vcc)	5.5 V	± 55 mV	P2
Power supply current limit, (Vcc)	1.18 A	NA	P1,P2
(V2)	-0.10 A	NA	P1,P2
(V3)	-0.23 A	NA	P1,P2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) The pattern voltages shall be applied as specified as follows:

Pin number	Pattern voltage				Test condition
	High (VH)	Accuracy	Low (VL)	Accuracy	
6,7,8,9,11,12, 13,14,15,16,17, 18,52,58,61,63, 64,65,66,67,68, 69,99	2.4 V	± 100 mV	0.4 V	± 100 mV	P1,P2
21,23,26,28,30, 34,36,38,43,45, 71,73,76,78,83, 85,87,89,92,94	-0.6 V	± 100 mV	-2.4 V	± 100 mV	P1,P2

(c) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

Pin number	Output compare levels				Test condition
	High level	Accuracy	Low level	Accuracy	
9,11,12,13, 15,16,17,18, 39,41,79,80	2.4 V ≤ Voh ≤ Vcc	± 105 mV	0.0 V ≤ Vol ≤ 0.5 V	± 25 mV	P1
9,11,12,13, 15,16,17,18, 39,41,79,80	2.4 V ≤ Voh ≤ Vcc	± 155 mV	0.0 V ≤ Vol ≤ 0.5 V	± 25 mV	P2
2,3,4,5,46, 47,48,49,53, 54,55,56,57, 59,62,95,96, 97,98,100	-0.6 V ≤ Voh ≤ 0.0 V	± 30 mV	-4.5 V ≤ Vol ≤ -2.5 V	± 100 mV	P1,P2

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NOTES: - Continued.

(d) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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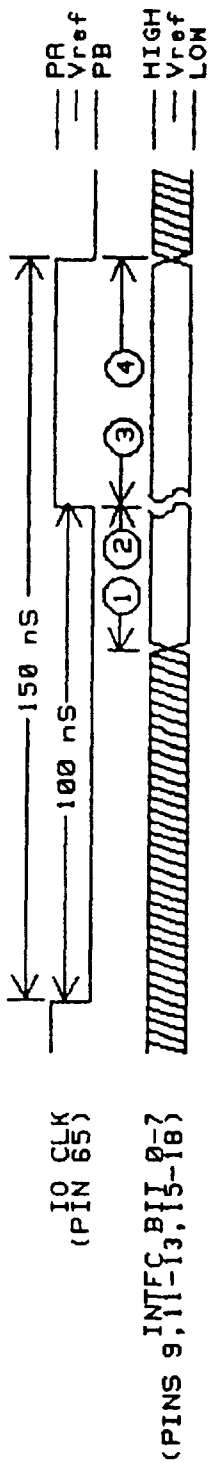


FIGURE 11. Input timing parameters for table XIV.

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NOTES:

1. (1) INTFC BIT 0-7 (CHAN N) to IO CLK. Minimum setup time: 15 nS.

2. (2) INTFC BIT 0-7 (CHAN N+1) to IO CLK. Minimum setup time: 15 nS.

3. (3) INTFC BIT 0-7 (CHAN N) to IO CLK. Minimum hold time: 10 nS.

4. (4) INTFC BIT 0-7 (CHAN N+1) to IO CLK. Minimum hold time: 10 nS.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (1000 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (1000 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (1000 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (1000 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Q1,Q3
(V2)	-5.2 V	± 55 mV	Q1-Q4
(V3)	-15.0 V	± 150 mV	Q1-Q4
(Vcc)	5.5 V	± 55 mV	Q2,Q4
Power supply current limit, (Vcc)	1.18 A	NA	Q1-Q4
(V2)	-0.10 A	NA	Q1-Q4
(V3)	-0.23 A	NA	Q1-Q4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The pattern voltages shall be applied as follows:

Pin number	Pattern voltage				Test condition
	High (VH)	Accuracy	Low (VL)	Accuracy	
6,7,8,9,11,12, 13,14,15,16,17, 18,52,58,61,63, 64,65,66,67,68, 69,99	2.4 V	± 100 mV	0.4 V	± 100 mV	Q1-Q4
21,23,26,28,30, 34,36,38,43,45, 71,73,76,78,83, 85,87,89,92,94	-0.6 V	± 100 mV	-2.4 V	± 100 mV	Q1-Q4

(c) The output compare levels shall be as follows:

Pin number	Output compare levels				Test condition
	High level	Accuracy	Low level	Accuracy	
9,11,12,13, 15,16,17,18, 39,41,79,80	Vref ≤ Voh ≤ Vcc	± 105 mV	0.0 V ≤ Vol ≤ Vref	± 25 mV	Q1,Q3
9,11,12,13, 15,16,17,18, 39,41,79,80	Vref ≤ Voh ≤ Vcc	± 155 mV	0.0 V ≤ Vol ≤ Vref	± 25 mV	Q2,Q4
2,3,4,5,46, 47,48,49,53, 54,55,56,57, 59,62,95,96, 97,98,100	Vref ≤ Voh ≤ 0.0 V	± 30 mV	-4.5 V ≤ Vol ≤ Vref	± 100 mV	Q1-Q4

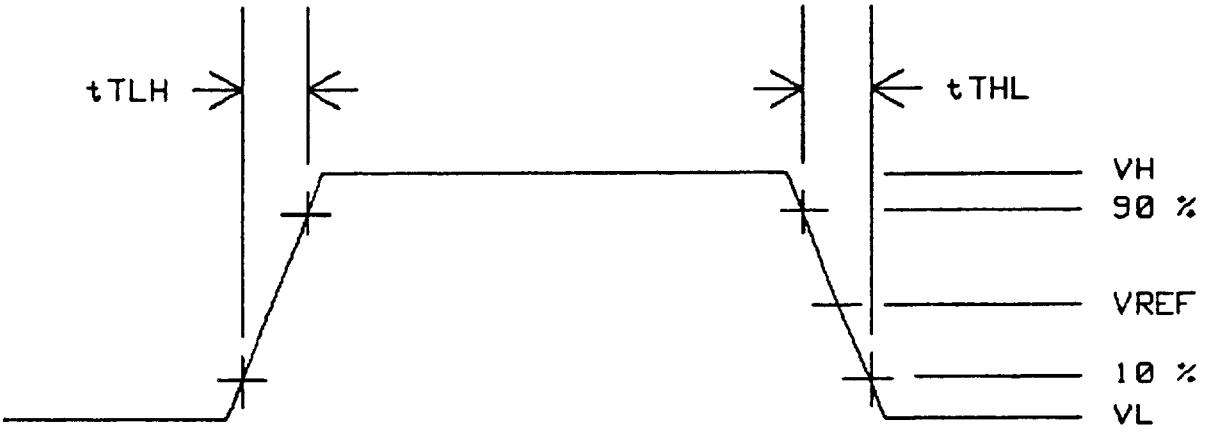
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NOTES: - Continued.

(d) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in note 2 (b).

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V for pins: 6,7,8,9,11,12,13,14,15,16,17,18, 39,41,52,58,61,63,64,65,66,67,68,69,79,80,99.

(d) Vref = -1.6 V for pins: 2,3,4,5,21,23,26,28,30,34,36, 38,43,45,46,47,48,49,53,54,55,56,57,59,62,71,73,76,78,83,85,87, 89,92,94,95,96,97,98,100.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1 1/	tsu	INTFC BIT 0 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 1 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 2 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 3 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 4 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 5 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 6 to IO CLK	15	15	15	-	-	-	nS	12
1	tsu	INTFC BIT 7 to IO CLK	15	15	15	-	-	-	nS	12
2 2/	tsu	INTFC BIT 0 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 1 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 2 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 3 to IO CLK	15	15	15	-	-	-	nS	12

1/ INTFC BIT 0 - INTFC BIT 7 is channel N.

2/ INTFC BIT 0 - INTFC BIT 7 is channel N+1.

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2 2/	tsu	INTFC BIT 4 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 5 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 6 to IO CLK	15	15	15	-	-	-	nS	12
2	tsu	INTFC BIT 7 to IO CLK	15	15	15	-	-	-	nS	12
3 1/	th	IO CLK to INTFC BIT 0	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 1	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 2	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 3	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 4	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 5	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 6	10	10	10	-	-	-	nS	12
3	th	IO CLK to INTFC BIT 7	10	10	10	-	-	-	nS	12
4 2/	th	IO CLK to INTFC BIT 0	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 1	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 2	10	10	10	-	-	-	nS	12

1/ INTFC BIT 0 - INTFC BIT 7 is channel N.

2/ INTFC BIT 0 - INTFC BIT 7 is channel N+1.

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4 2/	th	IO CLK to INTFC BIT 3	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 4	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 5	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 6	10	10	10	-	-	-	nS	12
4	th	IO CLK to INTFC BIT 7	10	10	10	-	-	-	nS	12
5	tpd	IO CLK to OUT 0 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 1 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 2 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 3 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 4 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 5 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 6 N	-	-	-	470	470	470	nS	12
5	tpd	IO CLK to OUT 7 N	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 0 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 1 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 2 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 3 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 4 N+1	-	-	-	470	470	470	nS	12

2/ INTFC BIT 0 - INTFC BIT 7 is channel N+1.

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	IO CLK to OUT 5 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 6 N+1	-	-	-	470	470	470	nS	12
6	tpd	IO CLK to OUT 7 N+1	-	-	-	470	470	470	nS	12
7	tpd	N BIT 0 to INTFC BIT 0	-	-	-	115	115	115	nS	12
7	tpd	N BIT 1 to INTFC BIT 1	-	-	-	115	115	115	nS	12
7	tpd	N BIT 2 to INTFC BIT 2	-	-	-	115	115	115	nS	12
7	tpd	N BIT 3 to INTFC BIT 3	-	-	-	115	115	115	nS	12
7	tpd	N BIT 4 to INTFC BIT 4	-	-	-	115	115	115	nS	12
7	tpd	N BIT 5 to INTFC BIT 5	-	-	-	115	115	115	nS	12
7	tpd	N BIT 6 to INTFC BIT 6	-	-	-	115	115	115	nS	12
7	tpd	N BIT 7 to INTFC BIT 7	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 0 to INTFC BIT 0	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 1 to INTFC BIT 1	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 2 to INTFC BIT 2	-	-	-	115	115	115	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	N+1 ID 3 to INTFC BIT 3	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 4 to INTFC BIT 4	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 5 to INTFC BIT 5	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 6 to INTFC BIT 6	-	-	-	115	115	115	nS	12
8	tpd	N+1 ID 7 to INTFC BIT 7	-	-	-	115	115	115	nS	12
9 1/	tpd	EI + LD C & CH N to INTFC BIT 0	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 1	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 2	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 3	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 4	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 5	-	-	-	65	65	65	nS	12
9	tpd	EI + LD C & CH N to INTFC BIT 6	-	-	-	65	65	65	nS	12

1/ INTFC BIT 0 - INTFC BIT 7 is channel N.

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9 1/	tpd	EI + LD C & CH N to INTFC BIT 7	-	-	-	65	65	65	nS	12
10 2/	tpd	EI + LD C & CH N to INTFC BIT 0	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 1	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 2	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 3	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 4	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 5	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 6	-	-	-	65	65	65	nS	12
10	tpd	EI + LD C & CH N to INTFC BIT 7	-	-	-	65	65	65	nS	12

1/ INTFC BIT 0 - INTFC BIT 7 is channel N.

2/ INTFC BIT 0 - INTFC BIT 7 is channel N+1.

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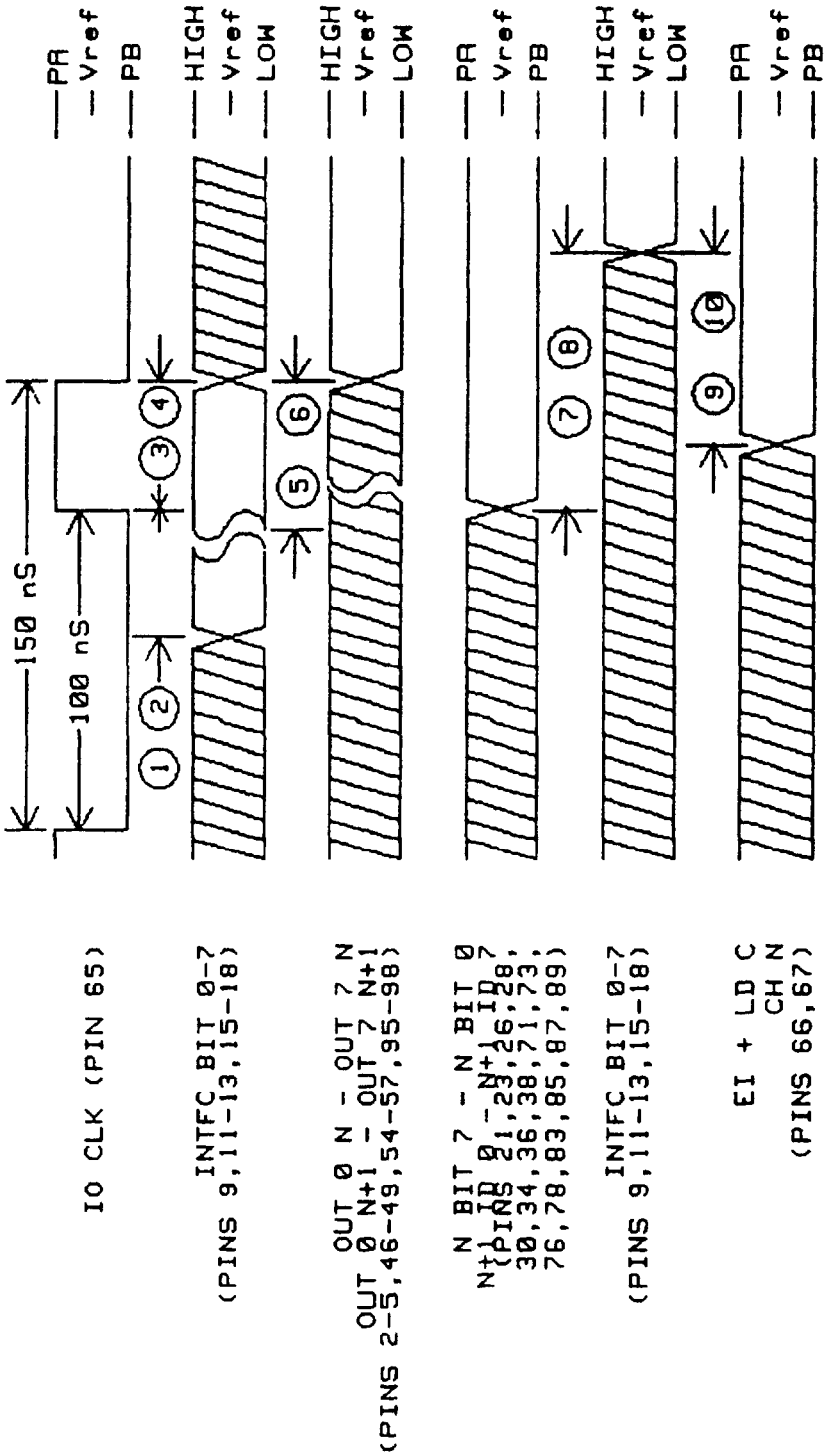


FIGURE 12. Dynamic waveforms for table XVI.

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TABLE XVII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	TEST N	6	2	75	pF	1-4
R2	LD EI RG N+1	7	2	100	pF	1-4
R3	FAST INTFC	8	8	200	pF	1-4
R4	LD ID RG N	14	2	75	pF	1-4
R5	N BIT 0 RTN	19	4	75	pF	1-4
R6	N BIT 0	21				
R7	N BIT 1 RTN	22				
R8	N BIT 1	23	4	75	pF	1-4
R9	N BIT 2 RTN	24	4	100	pF	1-4
R10	N BIT 2	26	4	175	pF	1-4
R11	N BIT 3 RTN	27	4	75	pF	1-4
R12	N BIT 3	28	4	100	pF	1-4
R13	N BIT 4 RTN	29				
R14	N BIT 4	30	4	100	pF	1-4
R15	N BIT 5 RTN	32	4	75	pF	1-4
R16	N BIT 5	34	4	100	pF	1-4
R17	N BIT 6 RTN	35	4	75	pF	1-4
R18	N BIT 6	36				
R19	N BIT 7 RTN	37				
R20	N BIT 7	38	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	ODR RTN N	42	4	75	pF	1-4
R22	ODR N (IN)	43				
R23	EFR RTN N	44				
R24	EFR N (IN)	45	4	75	pF	1-4
R25	ID ACK N	52	2	75	pF	1-4
R26	POWER UP	58	4	200	pF	1-4
R27	EF (A) CH N	61	2	50	pF	1-4
R28	OD (A) CH N	63	2	100	pF	1-4
R29	LD ID RG N+1	64	2	75	pF	1-4
R30	IO CLK	65	4	125	pF	1-4
R31	EI + LD C	66	4	100	pF	1-4
R32	CH N	67	4	100	pF	1-4
R33	RCV DATA	68	4	75	pF	1-4
R34	TEST N+1	69	2	75	pF	1-4
R35	N+1 ID 0 RTN	70	4	100	pF	1-4
R36	N+1 ID 0	71	4	100	pF	1-4
R37	N+1 ID 1 RTN	72	4	75	pF	1-4
R38	N+1 ID 1	73	4	100	pF	1-4
R39	N+1 ID 2 RTN	74	4	75	pF	1-4
R40	N+1 ID 2	76	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	N+1 ID 3 RTN	77	4	75	pF	1-4
R42	N+1 ID 3	78				
R43	N+1 ID 4 RTN	82				
R44	N+1 ID 4	83				
R45	N+1 ID 5 RTN	84				
R46	N+1 ID 5	85				
R47	N+1 ID 6 RTN	86				
R48	N+1 ID 6	87				
R49	N+1 ID 7 RTN	88				
R50	N+1 ID 7	89				
R51	EI RTN N	91				
R52	EI N (IN)	92				
R53	ID RTN N	93	4	75	pF	1-4
R54	ID N (IN)	94	4	100	pF	1-4
R55	EI ENBL N	99	2	50	pF	1-4
Bidirectionals						
R56	INTFC BIT 4	9	8	175	pF	1-4
R57	INTFC BIT 5	11				
R58	INTFC BIT 6	12				
R59	INTFC BIT 7	13				
R60	INTFC BIT 0	15	8	175	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	INTFC BIT 1	16	8	175	pF	1-4
R62	INTFC BIT 2	17	8	175	pF	1-4
R63	INTFC BIT 3	18	8	200	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R63
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R63
Capacitance measurement	NA	± 3.65 pF	R1
	NA	± 4.90 pF	R2
	NA	± 9.60 pF	R3
	NA	± 3.65 pF	R4
	NA	± 3.55 pF	R5-R8
	NA	± 4.80 pF	R9
	NA	± 8.55 pF	R10
	NA	± 3.55 pF	R11
	NA	± 4.80 pF	R12-R14
	NA	± 3.55 pF	R15
	NA	± 4.80 pF	R16
	NA	± 3.55 pF	R17-R24
	NA	± 3.65 pF	R25
	NA	± 9.80 pF	R26

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 2.40 pF	R27
	NA	± 4.90 pF	R28
	NA	± 3.65 pF	R29
	NA	± 6.05 pF	R30
	NA	± 4.80 pF	R31,R32
	NA	± 3.55 pF	R33
	NA	± 3.65 pF	R34
	NA	± 4.80 pF	R35,R36
	NA	± 3.55 pF	R37
	NA	± 4.80 pF	R38
	NA	± 3.55 pF	R39-R53
	NA	± 4.80 pF	R54
	NA	± 2.40 pF	R55
	NA	± 8.35 pF	R56-R62
	NA	± 9.6 pF	R63

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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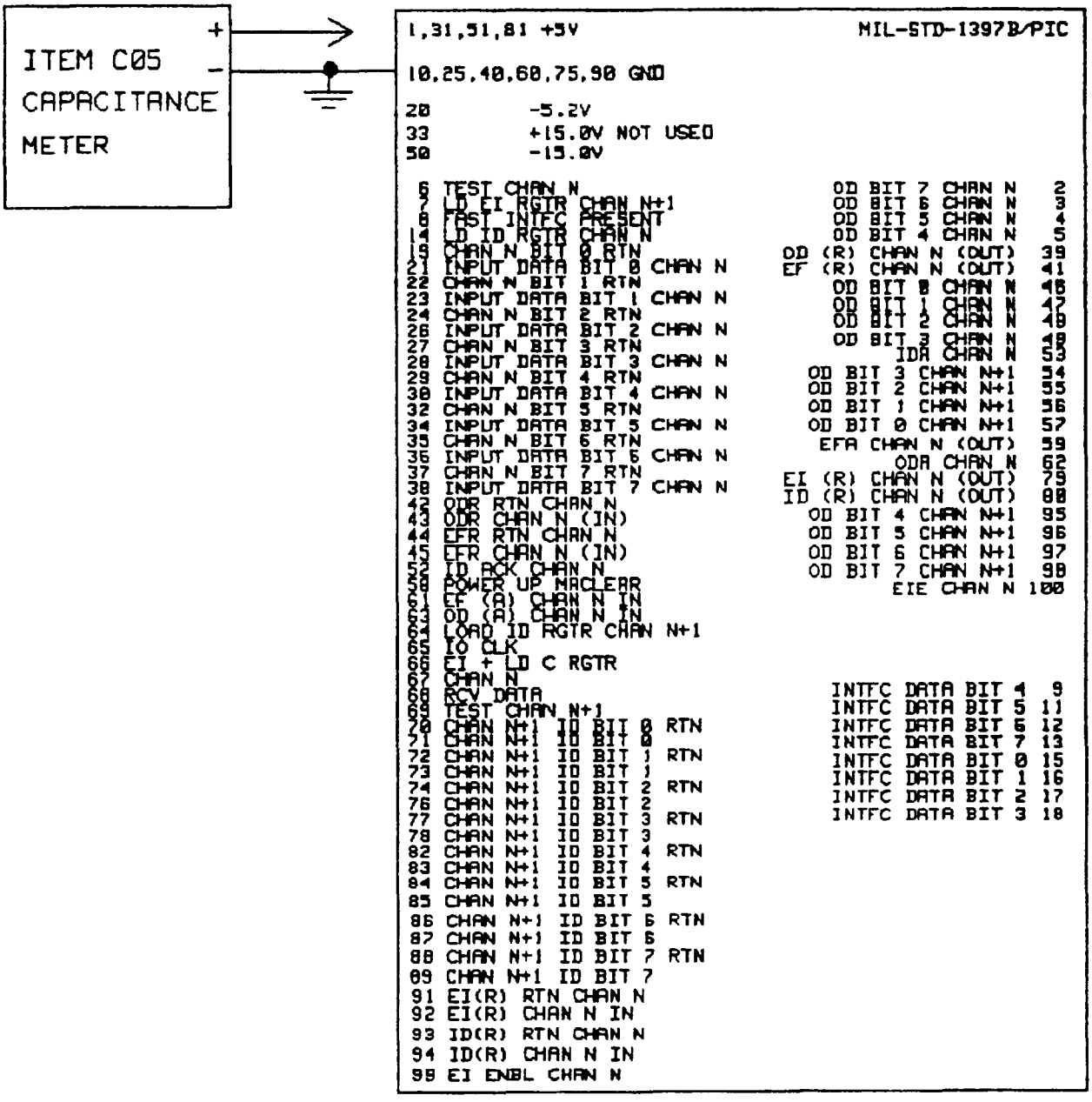


FIGURE 13. Input and output capacitance test circuit for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA, Vcc	1	0	400	mV	1-4
T2	PSTA, Vcc	51	0	400	mV	1-4
T3	PSTA, V2	20	0	400	mV	1-4
T4	PSTA, V3	50	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1-T4
(V2)	-5.2 V	± 250 mV	T1-T4
(V3)	-15.0 V	± 750 mV	T1-T4
Power supply current limit, (Vcc)	1.18 A	NA	T1-T4
(V2)	-0.10 A	NA	T1-T4
(V3)	-0.23 A	NA	T1-T4
Pattern voltage high, (VH)	3.0 V	± 100 mV	T1-T4
low, (VL)	0.0 V	± 100 mV	T1-T4
Power supply transient amplitude measurement	NA	± 20 mV	T1-T4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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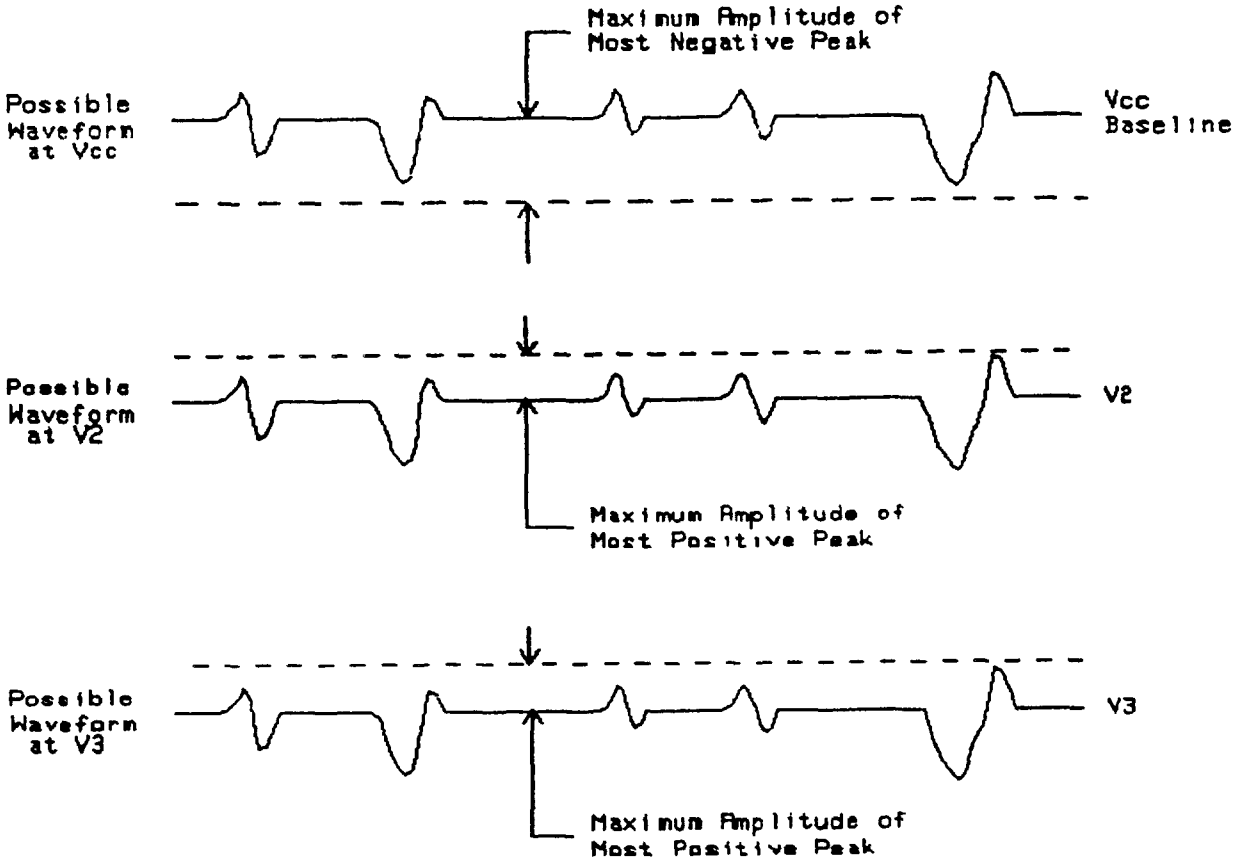
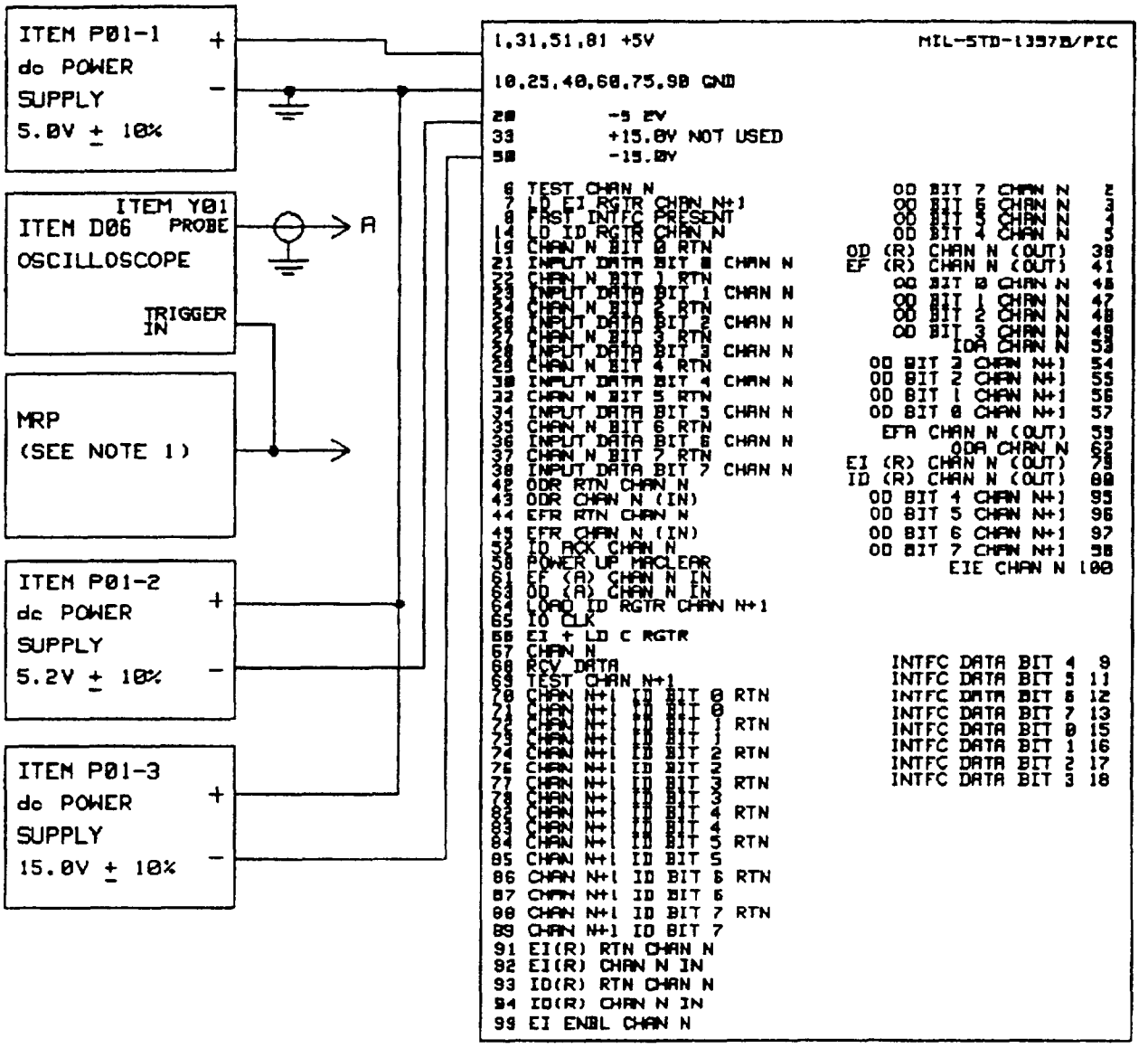


FIGURE 14. Power supply transient amplitude waveforms for table XVIII.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XVIII.

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